A Unified Framework for Convex Low-Rank SDP

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Abstract

In this paper, we develop a unified convex-optimization framework for low-rank semidefinite programming while avoiding the explicit nonconvex low-rank factor reformulation of the problem. Our framework provides a common perspective on conditional-gradient or Frank–Wolfe, spectral bundle, and low-rank projection methods. Building on this perspective, we develop two efficient low-dimensional convex quadratic-programming formulations within a common model-based primal–dual framework. A diagonal model provides lightweight spectral corrections with global sublinear convergence guarantees: $O(\epsilon^{-2})$ under general convexity and $O(\epsilon^{-1})$ under strict complementarity. The other, diagonally dominant (DD) model, introduces off-diagonal flexibility while retaining a low-dimensional convex QP structure. Under suitable local regularity conditions, we prove that the DD-QP locally reproduces the optimal correction of the full active-space SDP, even when the working dimension exceeds the optimal rank. This exactness allows us to establish local linear convergence without solving reduced semidefinite subproblems. We further extend the framework to heterogeneous multi-cone problems, combining exact treatment of inexpensive cone blocks with low-rank updates for large PSD blocks through a jointly coupled correction. The resulting hybrid algorithm uses diagonal corrections for inexpensive first-order progress and DD corrections for local linear acceleration, with the latter guarantee requiring a suitable primal–dual switching point. Our theoretical analyses and numerical experiments demonstrate the significant potential of this unified convex framework as a scalable alternative to the current nonconvex factor-based SDP approaches for large-scale semidefinite optimization.

1 Introduction

Semidefinite programming (SDP) [1, 2] is a fundamental modeling and relaxation tool in mathematical optimization. Its applications include graph optimization [3], circuit design [4], signal processing [5], robust portfolio optimization [6], quantum computing [7, 8], and electronic-structure calculations [9], just to mention a few. Interior-point methods have long been the principal approach to obtaining high-accuracy SDP solutions. Representative developments include primal–dual path-following methods [10] and dual-scaling methods such as DSDP [11], HDSDP [12], and PCDG [13]. Recent work on pure primal interior-point methods [14] also revisits the stability and computational potential of this approach in linear optimization. Their scalability, however, can still be limited by the matrix factorizations and coupled linear systems required at each iteration.

This has motivated considerable interest in algorithms that retain the convex formulation but use less expensive iterations. Examples include primal–dual splitting methods such as PDHG [15, 16] and SCS [17], augmented-Lagrangian methods such as Newton–CG ALM [18] and SDPNAL+ [19], and alternating-direction methods [20]. These approaches can be effective on large instances, but their full-space implementations may still require repeated PSD projections, large eigendecompositions, or coupled linear algebra, which become expensive for very large PSD blocks.

A natural way to alleviate the cost of repeated projections onto large PSD cones is to exploit low-rank spectral structure. In many SDP applications, the primal optimal matrix or the corresponding dual slack is

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observed to have low rank, or at least to exhibit rapidly decaying spectra [21], motivating algorithms that avoid full PSD projections at every iteration. Two broad approaches have emerged to exploit this structure. The classical Burer–Monteiro approach [22] parameterizes a PSD variable as $X = UU^\top$, substantially reducing the number of optimization variables but replacing the original convex SDP by a nonconvex problem, with the accompanying difficulty of certifying global optimality.

A different line of work preserves convexity and replaces full PSD projections by computations involving only a small amount of spectral information. Representative examples include Frank–Wolfe/conditional-gradient methods [23–25], which generate low-rank updates through leading eigenvectors; low-rank projection methods, including PDHG with low-rank rounding [16]; and spectral bundle methods [26, 27], which optimize richer convex models over adaptively selected spectral subspaces. We refer to this family broadly as *convex low-rank methods*, and it is the main focus of this paper.

Rank-one conditional-gradient methods provide the simplest convex low-rank strategy for SDP. On a trace-constrained PSD set, the Frank–Wolfe linear minimization oracle [24] reduces to a leading-eigenvector computation, so each iteration adds only one rank-one atom. Variants such as HCGM [28] and CGAL [25] extend this idea to problems with affine constraints through smoothing or augmented-Lagrangian updates. Their iterations remain inexpensive, but practical convergence often relies on diminishing step-size, smoothing, or penalty schedules. As a result, performance can be sensitive to parameter choices and progress may become slow when high-accuracy solutions are required.

Spectral bundle methods initiated by Helmberg and Rendl [26] and spectral Frank–Wolfe-type methods [29, 30] use richer spectral models by optimizing over several selected eigenvectors together with aggregate information. The convergence analysis of spectral bundle methods [27] shows that this additional flexibility can yield local linear convergence under suitable regularity conditions. Although GPU acceleration can make partial spectral computations increasingly efficient, richer models still require reduced semidefinite optimization. The bottleneck can therefore shift to the correction subproblem itself. In particular, for multi-block SDPs, the low-rank PSD blocks remain coupled through shared affine constraints, so a quadratic correction generally produces a joint quadratic SDP rather than independent small problems. Solving this coupled reduced SDP can therefore remain a significant computational bottleneck.

These complementary limitations motivate the following question: can the correction power of reduced semidefinite models be retained using convex QPs, without resorting to nonconvex factorization? Our contributions are threefold.

First, we develop a unified convex perspective on low-rank SDP algorithms by separating spectral subspace selection from the representation used within that subspace. This perspective provides a common way to describe rank-one conditional-gradient updates, spectral Frank–Wolfe and bundle corrections, and low-rank projection methods, while distinguishing their correction geometries and computational requirements. More importantly, it leads to two QP-based formulations for spectral corrections: a diagonal model that jointly optimizes the weights of selected spectral atoms and an aggregate matrix, and a primal-adapted diagonally dominant (DD) model that permits off-diagonal interactions in adapted coordinates. Both replace reduced PSD coefficient constraints by linear constraints, yielding convex QPs with $O(r)$ and $O(r^2)$ coefficients, respectively, for a working spectral dimension r.

Second, we establish local exactness of the primal-adapted DD correction. Under strict complementarity, uniqueness of the primal optimum, and suitable choices of the model parameters, we show that the DD-QP produces exactly the same optimal correction matrix as the full semidefinite problem on the active spectral subspace, provided that the current primal–dual state is sufficiently close to the optimal pair. This conclusion remains valid when the working dimension exceeds the optimal rank and does not require a spectral gap at the working dimension. The key is to show both that the active correction is representable in the adapted DD model and that the additional atoms cannot improve its objective. Thus, a polyhedral coefficient model can reproduce the full active-space SDP correction without solving that SDP. This exactness provides the connection needed to transfer spectral model-accuracy estimates to the DD update.

Third, we develop a hybrid diagonal–DD algorithm and establish convergence guarantees for its two phases. The diagonal method uses a mixed relative–absolute model-accuracy test and admits global

finite-accuracy bounds: averaged primal–dual errors reach ϵ in $O(\epsilon^{-2})$ QP solves under general convexity, while a current-center gap of ϵ in the exact dual penalty is attained in $O(\epsilon^{-1})$ QP solves under quadratic growth. Both bounds count rejected trials as well as accepted corrections. For a switch within the prescribed joint primal–dual neighborhood, the DD phase requires one QP per update and converges locally linearly, yielding an additional $O(\log(1/\epsilon))$ QP solves for dual distance ϵ. We further extend the construction and its two-phase analysis to heterogeneous multi-cone problems, where inexpensive cone blocks are represented exactly and large PSD blocks use low-rank spectral models. All blocks are optimized jointly through the shared affine constraints, yielding convex quadratic conic subproblems with simple projections in the retained block and spectral coefficient variables. The corresponding work bounds count exact joint model solves.

1.1 Other related work

Related convex low-rank methods. There are several complementary ways to exploit low-rank structure without factorizing the original problem. ProxSDP [31] approximates PSD projections by truncated spectral decompositions and adapts their rank. Low-rank extragradient methods [32] identify conditions under which locally computed low-rank projections recover the behavior of full projections. SketchyCGAL [33] and USBS [34] address storage through sketching, while approximate-complementarity methods [35] recover a low-rank primal solution from an accurate dual slack. These mechanisms address different parts of the computation and are complementary to the choice of correction model.

More directly related to our construction are spectral-atom and polyhedral bundle QPs [36], as well as structured LP/SOCP inner approximations [37, 38]. In particular, neither spectral-atom QPs nor basis-dependent DD approximations are new in themselves. Our results concern the specified diagonal and primal-adapted DD algorithms, the certificates that connect their corrections to the exact penalty, and their global and local guarantees. The distinction between subspace selection and within-subspace representation provides a common way to describe these developments.

GPU-accelerated low-rank SDP solvers. The nonconvex factorized approach has also benefited from recent GPU implementations. Building on LoRADS [39], cuLoRADS [40] uses an ADMM-based splitting to obtain GPU-friendly quadratic computations. Other developments include cuHALLaR [41], based on HALLaR [42], and the rank-adaptive method ALORA [43]. CARDAL [44] uses curvature information to target second-order stationarity at a fixed factor dimension before increasing the rank. These methods improve the computational viability of the Burer–Monteiro route while retaining its nonconvex factorized formulation. By contrast, the coefficient problems studied here remain convex because their spectral bases and congruence transformations are fixed during each correction.

Organization. The rest of the paper is organized as follows. Section 2 introduces the single-block framework, the five spectral models, and the main results for the diagonal and DD algorithms. Section 3 extends the construction to multi-cone problems. We then turn to the proofs. In Section 4, we establish the common primal–dual estimates under different model-accuracy conditions. Sections 5 and 6 verify these conditions for the diagonal and DD models and derive the corresponding work bounds. Section 7 presents the numerical studies, and Section 8 gives concluding remarks.

2 A unified framework for convex low-rank SDP

We begin with an SDP having a single PSD block. This setting allows us to separate the choice of spectral subspace from the choice of convex model without introducing block indices. After describing the common correction and its accuracy certificate, we present five realizations and state the main guarantees for the two QP-based algorithms. The multiple-cone extension will be developed in Section 3; its large PSD blocks use exactly the same models.

2.1 Problem setting

We consider the semidefinite program

$$\begin{aligned} \max_{X \in \mathbb{S}^n} \quad & \langle C, X \rangle \\ \text{s.t.} \quad & \mathcal{A}X = b, \\ & X \succeq 0, \end{aligned} \tag{2.1}$$

where $C \in \mathbb{S}^n$, $b \in \mathbb{R}^m$, and $\mathcal{A} : \mathbb{S}^n \to \mathbb{R}^m$ is linear. We equip $\mathbb{S}^n$ with the trace inner product and Frobenius norm, and $\mathbb{R}^m$ with the Euclidean norm. Let $\mathcal{A}^*$ be the adjoint of $\mathcal{A}$. We write $\|\mathcal{A}\|$ for the operator norm induced by these norms, and use the same convention for block operators. For a multiplier $y \in \mathbb{R}^m$, define

$$S(y) = C - \mathcal{A}^* y. \tag{2.2}$$

Thus $-S(y)$ is the usual dual slack. The dual problem is

$$\min_{y \in \mathbb{R}^m} \; b^\top y \qquad \text{s.t.} \qquad S(y) \preceq 0. \tag{2.3}$$

Standing assumptions. Throughout the single-block development, we assume that the primal and dual optimal values are finite, attained, and equal to f^*. Let X^* be a primal optimal solution, and suppose that a radius R satisfying $R > \operatorname{tr} X^*$ is known. We use this radius to define

$$\mathcal{D} = \{X \succeq 0 : \operatorname{tr} X \le R\}. \tag{2.4}$$

The trace restriction is not part of the original SDP. Its purpose is to provide a compact domain whose support function yields an exact penalty for the dual constraint. Specifically, define

$$F(y) = b^\top y + \max_{X \in \mathcal{D}} \langle S(y), X \rangle = b^\top y + R[\lambda_{\max}(S(y))]_+. \tag{2.5}$$

The spectral penalty [27] is exact under the standing assumptions: its minimum is f^*, and $\mathcal{Y}^* := \operatorname{argmin} F$ is precisely the dual optimal set. To see this, use the feasibility of X^* to obtain

$$F(y) - f^* = R[\lambda_{\max}(S(y))]_+ - \langle S(y), X^* \rangle \ge (R - \operatorname{tr} X^*)[\lambda_{\max}(S(y))]_+ \ge 0. \tag{2.6}$$

The strict inequality $R > \operatorname{tr} X^*$ implies that equality can hold only if $[\lambda_{\max}(S(y))]_+ = 0$ and $b^\top y = f^*$. Conversely, every dual optimum attains equality. Thus minimizing F is equivalent to solving the dual, although the iterates used to minimize it need not be dual feasible.

2.2 A common spectral correction interface

The central idea is to restrict each correction to a convex set described by a small number of spectral coefficients. In particular, low rank is exploited through the representation of the correction, not imposed as a nonconvex constraint. Given an aggregate $G \succeq 0$ and an orthonormal basis $V \in \mathbb{R}^{n \times r}$, a typical representation is

$$X = \beta G + VWV^\top, \qquad \beta \ge 0, \quad W \succeq 0. \tag{2.7}$$

During a correction, G and V are fixed. The map from (β, W) to X is therefore linear, and its coefficient constraints guarantee $X \succeq 0$. This is the essential distinction from optimizing over a factor in a representation such as $X = UU^\top$.

At a dual center y, we compute a correction by solving

$$\widehat{X} \in \operatorname*{argmin}_{X \in \mathcal{K}} \mathcal{J}_y(X), \tag{2.8}$$

where $\mathcal{K}$ is a closed convex spectral model and $\mathcal{J}_y$ is a convex objective. For the stabilized algorithms studied below, $\mathcal{K}$ is nonempty and contained in $\mathcal{D}$. The projection realization in Model 5 instead uses an

unbounded fixed-subspace cone with a Euclidean objective. Thus the interface specifies a common type of computation, not a single outer algorithm. The choices of spectral information, model, and update rule must all be specified. The resulting pattern is

spectral information $\longrightarrow$ convex spectral model $\longrightarrow$ primal correction $\longrightarrow$ optimality certificate.

We now specify the objective used in the diagonal and DD algorithms. For a fixed model $\mathcal{K} \subseteq \mathcal{D}$, center y, and parameter $\rho > 0$, set

$$\mathcal{J}_y(X) := \frac{\rho}{2}\|\mathcal{A}X - b\|^2 - \langle S(y), X\rangle. \tag{2.9}$$

This objective also appears in the augmented-Lagrangian corrections underlying CGAL [25] and spectral bundle methods [27]. Up to the constant $b^\top y$, it is the augmented Lagrangian for minimizing $-\langle C, X\rangle$. The quadratic term penalizes the equality residual. If y is dual feasible, the remaining term $\langle -S(y), X\rangle$ is nonnegative and measures complementarity; otherwise, it may have either sign. The associated dual trial is

$$\widehat{y} = y + \rho(\mathcal{A}\widehat{X} - b). \tag{2.10}$$

Model accuracy and acceptance. An exact minimizer over $\mathcal{K}$ need not give an accurate correction for the full domain $\mathcal{D}$. We therefore need to distinguish optimization over the current model from approximation of the original problem. The diagonal algorithm makes this distinction through an *a posteriori* model-error certificate. Unlike the prescribed step-size and penalty schedules used in CGAL [25], its acceptance decision depends on the accuracy of the computed trial.

For a nonempty compact set $\mathcal{K}$, write $\sigma_{\mathcal{K}}(H) := \max_{X\in\mathcal{K}}\langle H, X\rangle$ for its support function. The model $\mathcal{K} \subseteq \mathcal{D}$ induces the convex minorant

$$m(y) = b^\top y + \max_{X\in\mathcal{K}}\langle S(y), X\rangle \le F(y).$$

If $\widehat{X}$ solves the quadratic correction exactly and $\widehat{y}$ is given by (2.10), then Lemma 4.1 gives the support identity

$$m(\widehat{y}) = b^\top\widehat{y} + \langle S(\widehat{y}), \widehat{X}\rangle. \tag{2.11}$$

The same lemma gives the exact proximal representation

$$\widehat{y} = \operatorname*{argmin}_{\widetilde{y}\in\mathbb{R}^m}\left\{m(\widetilde{y}) + \frac{\|\widetilde{y} - y\|^2}{2\rho}\right\}. \tag{2.12}$$

Thus the correction performs proximal descent with the lower model m in place of F, which makes the accuracy of this model relevant to progress for the original objective. We then define

$$\Delta = F(y) - m(\widehat{y}), \qquad e = F(\widehat{y}) - m(\widehat{y}). \tag{2.13}$$

Here Δ is the predicted reduction, whereas $\Delta - e = F(y) - F(\widehat{y})$ is the actual reduction in the exact penalty. In particular, a trial satisfying $e \le \Delta/2$ realizes at least half of the predicted reduction. This comparison is analogous to the actual-versus-predicted reduction test in a trust-region method; no trust-region constraint is imposed. The diagonal algorithm also permits acceptance when the absolute model error is sufficiently small. By contrast, the DD algorithm takes its local corrections directly. Its analysis will show why an explicit acceptance loop is not needed in the prescribed neighborhood.

2.3 Five convex low-rank realizations

The following five models make the tradeoff between correction flexibility and subproblem cost explicit. Models 1, 4, and 5 describe connections to single-atom updates, full spectral-subspace corrections, and low-rank projections, respectively. Models 2 and 3 are the QP realizations analyzed in this paper. As discussed in the introduction, spectral-atom QPs and basis-dependent DD approximations have precedents [36, 38]; the new analysis concerns the algorithms and guarantees developed from the particular constructions below.

2.3.1 Model 1: single-atom correction

We first retain only an aggregate $G \in \mathcal{D}$ and one new atom. Given a symmetric spectral source Ξ, choose a unit leading eigenvector v, and set $A_v = Rvv^\top$ if $\lambda_{\max}(\Xi) > 0$ and $A_v = 0$ otherwise. The correction model is

$$\mathcal{K}^{(1)}(G, v) = \{(1-\alpha)G + \alpha A_v : 0 \le \alpha \le 1\}. \tag{2.14}$$

For a smooth convex objective f, taking $\Xi = -\nabla f(G)$ and minimizing f on this segment recovers a Frank–Wolfe update with exact line search [24]. For the quadratic objective (2.9), the corresponding source is

$$\Xi := -\nabla \mathcal{J}_y(G) = S(y) - \rho \mathcal{A}^*(\mathcal{A}G - b). \tag{2.15}$$

Thus the augmented-Lagrangian correction has the same primal conditional-gradient mechanism as CGAL [25]. This connection does not include CGAL's multiplier, penalty, or step-size rules: a common correction objective does not make the outer methods identical.

2.3.2 Model 2: diagonal spectral QP

A natural extension is to retain several spectral directions and optimize their weights together with the aggregate coefficient. For $G \in \mathcal{D}$, choose an orthonormal leading eigenframe $V = [v_1, \dots, v_r] \in \mathrm{TopEig}_r(S(y))$ and define

$$\begin{aligned} \mathcal{K}^{(2)}(G, V) &= \mathrm{conv}\left\{0,\ G,\ Rv_1v_1^\top, \dots, Rv_rv_r^\top\right\} \\ &= \left\{\beta G + RV\,\mathrm{Diag}(w)V^\top : \beta \ge 0,\ w \ge 0,\ \beta + \mathbf{1}^\top w \le 1\right\}. \end{aligned} \tag{2.16}$$

Using (2.9) on this model gives a convex QP in $r+1$ scalar coefficients with only linear constraints. Unlike a single-atom update, the correction reoptimizes all retained weights simultaneously. These weights are decision variables, rather than the eigenvalues of the matrix from which the directions were selected.

The term *diagonal* describes the new coefficient matrix $\mathrm{Diag}(w)$, not the full matrix X. In particular, the aggregate G need not be diagonal in the basis V, and it retains information from earlier corrections. The model therefore offers more flexibility than the rank-one segment while keeping the coefficient problem especially small.

2.3.3 Model 3: primal-adapted DD-QP

The diagonal model does not allow interactions between newly selected directions. To introduce such interactions without restoring a PSD coefficient constraint, we use the diagonally dominant cone

$$\mathsf{DD}_r := \left\{T \in \mathbb{S}^r : T_{aa} \ge \sum_{b \ne a} |T_{ab}|, \quad a = 1, \dots, r\right\} \subseteq \mathbb{S}^r_+. \tag{2.17}$$

This cone is polyhedral and permits off-diagonal coefficients. Its inclusion in $\mathbb{S}^r_+$ follows from the following consequence of Gershgorin's theorem [45, Section 6.1].

Fact 2.1. *If $T \in \mathbb{S}^r$ satisfies $T_{aa} - \sum_{b \ne a} |T_{ab}| \ge c$ for every a, then $T \succeq cI$.*

The choice of coordinates is important because diagonal dominance is basis-dependent. To adapt the coordinates to $G \in \mathcal{D}$, let $V \in \mathrm{St}(n, r)$ be an ordered leading eigenframe of $S(y)$, and choose the upper triangular factor with positive diagonal satisfying

$$V^\top G V + \varepsilon I_r = LL^\top, \qquad \varepsilon > 0. \tag{2.18}$$

Here $\mathrm{St}(n, r) = \{V \in \mathbb{R}^{n \times r} : V^\top V = I_r\}$. The adapted model is

$$\mathcal{K}^{(3)}(V, L) = \left\{VLTL^\top V^\top : T \in \mathsf{DD}_r,\ \mathrm{tr}(LTL^\top) \le R\right\}. \tag{2.19}$$

The transformation is fixed during each correction, so the model remains convex. Since VL need not have orthonormal columns, the trace constraint must be imposed on the lifted matrix rather than on T alone.

Using (2.9) on this model gives a convex QP with $O(r^2)$ variables and simplex constraints, as detailed in Section 2.5. The triangular choice also preserves the leading eigenvector, since $Le_1 = L_{11}e_1$; no separate center-supporting atom is needed. There is, however, an important difference from Model 2: the aggregate G determines the coordinates but is not itself retained as an independent atom. The direct DD iteration will therefore be analyzed under a joint local primal–dual initialization, not an arbitrary one.

2.3.4 Model 4: full reduced semidefinite correction

For comparison, allowing a full PSD coefficient matrix gives the model

$$\mathcal{K}^{(4)}(G,V) = \left\{\beta G + VWV^\top : \beta \geq 0,\ W \succeq 0,\ \beta + \frac{\operatorname{tr} W}{R} \leq 1\right\}. \tag{2.20}$$

Related aggregate-plus-PSD representations appear in spectral Frank–Wolfe and kFW methods [29, 30], with trace-equality normalization, and in the proximal support models of spectral bundle methods [26, 27]. Again, the common feature is the correction geometry; the outer update rules differ.

With objective (2.9), this model produces a quadratic SDP with $r(r+1)/2$ symmetric coefficients and one aggregate coefficient. Models 2 and 3 replace this PSD constraint by polyhedral constraints of different expressive power. In the local DD analysis, the full reduced model with $\beta = 0$ will serve as a comparison problem on the active subspace. That auxiliary SDP is used in the proof only; it is not solved by the DD algorithm.

2.3.5 Model 5: low-rank proximal projection

The final realization concerns a different correction objective. Given a proximal input $Z = X + \tau S(y)$, let $V = [v_1, \ldots, v_r]$ contain its r leading eigenvectors and define the fixed-subspace cone

$$\mathcal{K}^{(5)}(V) = \{VWV^\top : W \succeq 0\}. \tag{2.21}$$

The Euclidean correction is explicit:

$$X^+ = \operatorname*{argmin}_{X \in \mathcal{K}^{(5)}(V)} \frac{1}{2}\|X - Z\|_F^2 = V(V^\top ZV)_+ V^\top = \sum_{a=1}^{r} [\lambda_a(Z)]_+ v_a v_a^\top. \tag{2.22}$$

For exact eigenpairs ordered by algebraic value,

$$\|X^+ - \Pi_{\mathbb{S}^n_+}(Z)\|_F^2 = \sum_{a>r} [\lambda_a(Z)]_+^2. \tag{2.23}$$

This is the approximate-projection mechanism used in ProxSDP-type splitting methods [31]. Its error is controlled by the omitted positive spectrum. Unlike Models 2–4, it minimizes Euclidean distance and imposes no trace cap; adding such a cap changes the projection formula. Moreover, convergence of the resulting splitting method requires its own projection-error conditions and does not follow from the support-model analysis below. Table 1 summarizes the five realizations.

2.4 A single-block diagonal-QP algorithm

We now use Model 2 to construct the algorithm in Algorithm 2.1. There are two levels of iteration. The outer index k counts accepted center updates; the inner index i counts refinements while the center y_k is held fixed. Starting with $X_{k,0} = X_k$ and $y_{k,0} = y_k$, choose $V_{k,i} \in \mathbb{R}^{n\times r}$ to contain the top-r eigenvectors of $S(y_{k,i})$, ordered by algebraic eigenvalue. The current model is

$$\mathcal{K}_{k,i} = \left\{\beta X_{k,i} + RV_{k,i}\operatorname{Diag}(w)V_{k,i}^\top : \beta \geq 0,\ w \geq 0,\ \beta + \mathbf{1}^\top w \leq 1\right\}. \tag{2.24}$$

Table 1: Five single-block realizations. Sizes count reduced variables, not arithmetic operations.

	Model	Reduced computation	Role / connection
1	Single atom	One weight	FW / CGAL correction
2	Diagonal	$r+1$ coefficients; QP	Low-cost realization
3	Adapted DD	$O(r^2)$ variables; QP	Local acceleration
4	Full spectral	$O(r^2)$ variables; SDP	Spectral FW / bundle
5	Low-rank projection	Truncated positive spectrum	ProxSDP-type splitting

Thus the exact correction reduces to the coefficient QP

$$(\beta_{k,i}, w_{k,i}) \in \operatorname*{argmin}_{\beta\geq 0,\, w\geq 0,\, \beta+\mathbf{1}^\top w\leq 1} \mathcal{J}_{y_k}\left(\beta X_{k,i} + RV_{k,i}\operatorname{Diag}(w)V_{k,i}^\top\right), \tag{2.25}$$

which has $r+1$ variables and only linear constraints. The resulting primal correction and its associated dual trial are

$$\begin{aligned} X_{k,i+1} &= \beta_{k,i}X_{k,i} + RV_{k,i}\operatorname{Diag}(w_{k,i})V_{k,i}^\top, \\ y_{k,i+1} &= y_k + \rho(\mathcal{A}X_{k,i+1} - b). \end{aligned} \tag{2.26}$$

It is important that the center in both $\mathcal{J}_{y_k}$ and the dual trial formula remains y_k. Thus the inner loop does not apply successive dual updates. Instead, it constructs a succession of trials for the same center by changing the aggregate and the spectral model.

For $m_{k,i}(y) = b^\top y + \sigma_{\mathcal{K}_{k,i}}(S(y))$, the predicted reduction and model error are

$$\Delta_{k,i} = F(y_k) - m_{k,i}(y_{k,i+1}), \qquad e_{k,i} = F(y_{k,i+1}) - m_{k,i}(y_{k,i+1}). \tag{2.27}$$

The trial is accepted if the model error is sufficiently small relative to the predicted progress or meets the prescribed absolute model accuracy:

$$e_{k,i} \leq \frac{\Delta_{k,i}}{2} \qquad \text{or} \qquad e_{k,i} \leq \eta, \tag{2.28}$$

where $\eta \geq 0$ is a prescribed model-accuracy tolerance.

A rejected trial leaves y_k unchanged, but is not discarded. Its primal output $X_{k,i+1}$ becomes the next aggregate, and the leading eigenvectors of $S(y_{k,i+1})$ supply the new directions. The next model therefore contains the preceding correction and r new spectral atoms; together with zero, these include an exact support atom. This retention property, rather than nestedness of the models, is what allows the refinement cost to be bounded. The leading-eigenvalue computation used to evaluate $F(y_{k,i+1})$ can also produce the top-r directions for the next refinement.

Our first convergence result addresses two questions: whether an acceptable trial can be obtained after finitely many refinements, and how the accepted outputs approximate a primal–dual solution. To state it, choose an optimum $y^{\mathrm{ref}} \in \mathcal{Y}^*$ nearest to y_0 and define

$$\begin{aligned} &h_0 := F(y_0) - f^*, \qquad d_0 := \operatorname{dist}(y_0, \mathcal{Y}^*), \qquad \Gamma := h_0 + \frac{(\|y_0\| + 2\|y^{\mathrm{ref}}\| + 1)^2}{2\rho}, \\ &C := \rho D_{\mathcal{A}}^2, \qquad D_{\mathcal{A}} := \sup_{X,Z\in\mathcal{D}} \|\mathcal{A}(X-Z)\| \leq 2R\|\mathcal{A}\|. \end{aligned} \tag{2.29}$$

The work unit is one exact diagonal-QP solve, including a solve whose trial is subsequently rejected. The numerical iterations used to solve a coefficient QP are not included in this count.

Theorem 2.2 (Diagonal QP under general convexity)**.** *Under the standing assumptions, run Algorithm 2.1 with a fixed tolerance $\eta > 0$. Then every inner loop terminates after at most $2 + 5C/\eta$ QP solves. Let X_k*

Algorithm 2.1 Single-block diagonal spectral QP

Input: $X_0 \in \mathcal{D}$, $y_0 \in \mathbb{R}^m$, $1 \le r \le n$, $\rho > 0$, and $\eta \ge 0$.

for $k = 0, 1, 2, \ldots$ **do**
  Set $X_{k,0} = X_k$ and $y_{k,0} = y_k$.
  for $i = 0, 1, 2, \ldots$ **do**
    Compute $V_{k,i} = \mathrm{TopEig}_r(S(y_{k,i}))$.
    Solve (2.25) for $(\beta_{k,i}, w_{k,i})$.
    Form $(X_{k,i+1}, y_{k,i+1})$ from (2.26).
    Compute $(\Delta_{k,i}, e_{k,i})$ from (2.27).
    if $e_{k,i} \le \Delta_{k,i}/2$ or $e_{k,i} \le \eta$ **then**
      Set $X_{k+1} = X_{k,i+1}$ and $y_{k+1} = y_{k,i+1}$.
      break
    end if
  end for
end for

and y_k denote the accepted primal and dual iterates, and define $\bar{X}_K = \frac{1}{K}\sum_{k=1}^{K} X_k$ and $\bar{y}_K = \frac{1}{K}\sum_{k=1}^{K} y_k$. Then $\bar{X}_K \in \mathcal{D}$ for every $K \ge 1$ and

$$\max\left\{F(\bar{y}_K) - f^*, \|\mathcal{A}\bar{X}_K - b\|, \left|\langle C, \bar{X}_K\rangle - f^*\right|\right\} \le \frac{\Gamma}{K} + 2\eta. \tag{2.30}$$

Consequently, for any target accuracy $\epsilon > 0$, choosing $\eta = \frac{\epsilon}{4}$ and $K = \max\left\{1, \left\lceil \frac{2\Gamma}{\epsilon} \right\rceil\right\}$ makes all three errors in (2.30) at most ϵ, and the total number of diagonal-QP solves is bounded by

$$N_{\mathrm{QP}} \le K\left(2 + \frac{20C}{\epsilon}\right) = O(\epsilon^{-2}). \tag{2.31}$$

Theorem 2.2 gives a guarantee for averaged accepted outputs without a growth assumption. Its proof combines the common primal–dual comparison in Section 4 with a fixed-center refinement bound in Section 5.1. The use of averages is important: this result does not assert the same bounds for the current raw primal matrix.

We next ask whether additional geometry can improve the current-center guarantee. The property we use is quadratic growth of the exact penalty. To account for redundant equalities, let $\mathcal{E}_0 := y_0 + \mathrm{range}\,\mathcal{A}$ and $\mathcal{L}_H := \{y : F(y) - f^* \le H\}$. Primal feasibility gives $b = \mathcal{A}X^*$, so every dual update stays in $\mathcal{E}_0$. The following assumption provides a source of quadratic growth.

Assumption 2.1 (Strict complementarity)**.** Under the standing assumptions, there is an optimal pair $(X_\star, y_\star)$ such that, with $Z_\star := -S(y_\star) = \mathcal{A}^* y_\star - C$, we have $X_\star Z_\star = 0$, $X_\star + Z_\star \succ 0$, and $R > \mathrm{tr}\, X_\star$.

Assumption 2.2 (Quadratic growth)**.** We distinguish the following two growth conditions.

(i) For the relevant sublevel $\mathcal{E}_0 \cap \mathcal{L}_H$, there is $\mu > 0$ such that

$$F(y) - f^* \ge \frac{\mu}{2}\,\mathrm{dist}^2(y, \mathcal{Y}^*), \qquad y \in \mathcal{E}_0 \cap \mathcal{L}_H. \tag{2.32}$$

(ii) There are $\mu > 0$ and $r_{\mathrm{QG}} > 0$ such that

$$F(y) - f^* \ge \frac{\mu}{2}\,\mathrm{dist}^2(y, \mathcal{Y}^*), \qquad y \in B(y_\star, r_{\mathrm{QG}}). \tag{2.33}$$

The moduli in these two conditions may differ. The diagonal analysis uses (i), whereas the local DD analysis uses (ii).

Remark. After removing redundant equalities, matching the objective sign, and eliminating the dual slack, the exact-penalty result in [46, Theorem 2] gives quadratic growth on compact sets under Assumption 2.1. Consequently, Assumption 2.1 implies Assumption 2.2(ii). If $\mathcal{E}_0 \cap \mathcal{L}_H$ is bounded, it also implies Assumption 2.2(i). A sufficient condition for bounded relevant sublevels, based on a strictly feasible primal point, is given in Appendix A.

The next theorem states the improved guarantee in terms of this growth property. For a given modulus $\mu > 0$, define

$$c := \min\left\{\frac{1}{2}, \frac{\rho\mu}{4}\right\}, \quad q := \frac{1+\rho\mu}{1+2\rho\mu} < 1, \quad C_0 := h_0 + \frac{d_0^2}{2\rho}, \quad B := \frac{2(1+\rho\mu)}{\rho\mu}. \tag{2.34}$$

Theorem 2.3 (Diagonal QP under quadratic growth)**.** *Suppose the standing assumptions hold, $h_0 \le H$ for some $H > 0$, and Assumption 2.2(i) holds with $\mu > 0$. Run Algorithm 2.1 with $0 < \eta \le cH/2$. Then all accepted centers remain in $\mathcal{E}_0 \cap \mathcal{L}_H$, and, for every $K \ge 1$,*

$$F(y_K) - f^* \le C_0 q^K + B\eta, \qquad \operatorname{dist}^2(y_K, \mathcal{Y}^*) \le 2\rho\big(C_0 q^K + B\eta\big). \tag{2.35}$$

For the corresponding raw primal output, define $r_K^{\mathrm{p}} := \sqrt{\frac{2}{\rho}\left(C_0 q^{K-1} + (B+1)\eta\right)} = O\big(q^{K/2} + \sqrt{\eta}\big)$. Then

$$\|\mathcal{A}X_K - b\| \le r_K^{\mathrm{p}}. \tag{2.36}$$

If, in addition, $M_H := \sup_{y \in \mathcal{E}_0 \cap \mathcal{L}_H} \|y\| < \infty$, then the raw primal objective satisfies

$$\big|\langle C, X_K\rangle - f^*\big| \le C_0 q^{K-1} + (B+2)\eta + M_H r_K^{\mathrm{p}}. \tag{2.37}$$

For all sufficiently small $\epsilon > 0$, a suitable fixed $\eta = \Theta(\epsilon)$ produces, within $O(\epsilon^{-1})$ diagonal-QP solves, an accepted primal–dual pair with dual penalty gap at most ϵ and dual-distance, raw primal feasibility, and raw primal objective errors of order $\sqrt{\epsilon}$. This finite-target guarantee does not require $M_H < \infty$.

There are two distinctions worth emphasizing. First, quadratic growth improves the estimate for the *current* dual center, whereas the raw primal errors have a square-root dependence on the model tolerance. The additional bound $M_H < \infty$ is needed only for the uniform absolute objective estimate of the raw primal outputs. Once quadratic growth on the relevant set is available, the dual bounds and primal feasibility bound use only that property and sublevel invariance. A uniform bound on the accepted multipliers can replace sublevel boundedness in the raw objective argument.

Second, the factor $q < 1$ concerns accepted center updates, not the total number of QP solves. The refinement cost must still be counted. This distinction leads to the finite-accuracy work bounds $O(\epsilon^{-2})$ and $O(\epsilon^{-1})$ in the two theorems. For a fixed positive tolerance, the estimates retain an error floor; they are not zero-error convergence statements. At $\eta = 0$, only the relative test is used, and the inner loop need not terminate finitely at an already optimal center. The proofs of both diagonal theorems are given in Section 5.

2.5 A single-block local DD-QP algorithm

The diagonal algorithm controls model accuracy by refinement. We now consider a different mechanism: enrich the within-subspace model so that the desired accuracy follows locally from its geometry. The DD algorithm uses a fixed working dimension r, adapts the DD coordinates to the preceding primal matrix, and takes every correction directly. The purpose of the upper triangular transformation is to preserve the ordered leading subspaces while changing the coefficient cone.

For $V \in \mathrm{St}(n, r)$ and an upper triangular matrix L with positive diagonal, define

$$\mathcal{K}(V, L) := \left\{VLTL^\top V^\top : T \in \mathsf{DD}_r,\ \operatorname{tr}(LTL^\top) \le R\right\} \subseteq \mathcal{D}. \tag{2.38}$$

The factor order matters: throughout, L is upper triangular and appears as $LL^\top$, rather than $L^\top L$. For $V \in \mathrm{St}(n, r)$ and $1 \le j \le r$, we have

$$(VL)_{:,1:j} = V_{:,1:j}L_{1:j,1:j}, \qquad \operatorname{range}((VL)_{:,1:j}) = \operatorname{range}(V_{:,1:j}). \tag{2.39}$$

Thus each leading subspace is preserved. To obtain an explicit QP representation, we use the extreme-ray characterization of the DD cone [47, Theorem 2]; see also [38, Lemma 3.2]. For the standard basis vectors $e_a \in \mathbb{R}^r$, set

$$\mathcal{V}_r = \{e_a : 1 \le a \le r\} \cup \left\{(e_a + e_b)/\sqrt{2}, (e_a - e_b)/\sqrt{2} : 1 \le a < b \le r\right\}. \tag{2.40}$$

These r^2 directions give the equivalent representations

$$\mathsf{DD}_r = \left\{ \sum_{z \in \mathcal{V}_r} \alpha_z z z^\top : \alpha_z \geq 0 \right\}, \tag{2.41}$$

$$\{T \in \mathsf{DD}_r : \operatorname{tr} T \leq R\} = \operatorname{conv}\big(\{0\} \cup \{R z z^\top : z \in \mathcal{V}_r\}\big). \tag{2.42}$$

Normalizing the transformed directions VLz gives an analogous convex hull of zero and r^2 trace-R atoms.

Proposition 2.4. *For $\mathcal{K}(V, L)$ in (2.38), let $a_z := VLz/\|VLz\|$ and $A_z := R a_z a_z^\top$ for $z \in \mathcal{V}_r$. Then*

$$\mathcal{K}(V, L) = \operatorname{conv}\big(\{0\} \cup \{A_z : z \in \mathcal{V}_r\}\big) \subseteq \mathcal{D}. \tag{2.43}$$

Proof. We prove the two inclusions separately. Suppose first that $X = \sum_z \alpha_z A_z$ with $\alpha_z \geq 0$ and $\sum_z \alpha_z \leq 1$. Set $T = \sum_z R\alpha_z z z^\top / \|VLz\|^2$. Then $T \in \mathsf{DD}_r$, $VLTL^\top V^\top = X$, and $\operatorname{tr}(LTL^\top) = R\sum_z \alpha_z \leq R$. Conversely, express a feasible T as $\sum_z \beta_z z z^\top$ using (2.41). Its lift is $\sum_z (\beta_z \|VLz\|^2/R) A_z$, and the sum of these coefficients is $\operatorname{tr}(LTL^\top)/R \leq 1$. This proves the reverse inclusion. Finally, every atom is PSD with trace R, so the convex hull is contained in $\mathcal{D}$. □

For $Z \in \mathbb{S}^r_{++}$, we use $\operatorname{chol}(Z)$ to denote the unique upper triangular matrix L with positive diagonal such that $Z = LL^\top$. This is the reverse Cholesky convention [48, Theorem 6(ii) and Remark 1]. The factor depends continuously on Z, as follows from the smoothness of the ordinary Cholesky map on the positive definite cone [49, Proposition 2]. This continuity will allow the local DD representation to persist when the primal matrix and spectral frame are perturbed.

Given V_k and L_k, the primal correction is

$$X_{k+1} \in \operatorname*{argmin}_{X \in \mathcal{K}(V_k, L_k)} \mathcal{J}_{y_k}(X). \tag{2.44}$$

The dual center is then updated directly. The complete iteration is given in Algorithm 2.2.

Algorithm 2.2 Single-block local upper-triangular DD-QP

Input: $y_0 \in \mathbb{R}^m$, $X_0 \in \mathcal{D}$, working dimension $1 \leq r \leq n$, $\rho > 0$, and $\varepsilon > 0$.

1: **for** $k = 0, 1, 2, \ldots$ **do**
2: Compute an ordered top-r spectral frame $V_k \in \operatorname{TopEig}_r(S(y_k))$.
3: Set $L_k = \operatorname{chol}(V_k^\top X_k V_k + \varepsilon I_r)$.
4: Solve the DD correction (2.44) for X_{k+1}.
5: Set $y_{k+1} = y_k + \rho(\mathcal{A}X_{k+1} - b)$.
6: **end for**

The columns of V_k are ordered by nonincreasing eigenvalue; any orthonormal basis may be chosen within a repeated eigenspace. Each iteration consists of one top-r spectral computation and one convex QP. The algorithm does not take the optimal rank as an input. Its local guarantee permits any fixed $r \geq s$ and does not assume a gap at the working rank r. The distinction between this working dimension and the optimal rank is central to the exactness argument below.

Coefficient-QP representation. Enumerate the directions in (2.40) as $z_1, \ldots, z_{r^2}$. Proposition 2.4 gives

$$\mathcal{K}(V_k, L_k) = \left\{ \sum_{\ell=1}^{r^2} \alpha_\ell A_{k,\ell} : \alpha \geq 0,\ \mathbf{1}^\top \alpha \leq 1 \right\} \quad \text{where} \quad a_{k,\ell} := \frac{V_k L_k z_\ell}{\|V_k L_k z_\ell\|},\ A_{k,\ell} := R\, a_{k,\ell} a_{k,\ell}^\top. \tag{2.45}$$

Let D_k have columns $\mathcal{A}(A_{k,\ell})$ and define $(\ell_k)_\ell = \langle S(y_k), A_{k,\ell} \rangle$. Then (2.44) is equivalently

$$X_{k+1} = \sum_{\ell=1}^{r^2} (\alpha_k)_\ell A_{k,\ell} \quad \text{where} \quad \alpha_k \in \operatorname*{argmin}_{\alpha \geq 0,\ \mathbf{1}^\top \alpha \leq 1} \left\{ \frac{\rho}{2} \|D_k \alpha - b\|^2 - \ell_k^\top \alpha \right\}. \tag{2.46}$$

The Hessian is $\rho D_k^\top D_k \succeq 0$, and the constraints form a simplex-type polytope. Thus no reduced PSD constraint is present. The atomic coefficients need not be unique, even when the matrix that they represent is unique.

For the local analysis, let $s = \operatorname{rank} X_\star$, $\sigma_\star = \lambda_s(X_\star) > 0$, and let $\gamma_\star > 0$ denote the inactive dual-slack spectral gap implied by strict complementarity. Define

$$L_{\mathrm{sp}} := \frac{2R\|\mathcal{A}\|^2}{\gamma_\star}. \tag{2.47}$$

The local model-error argument in Lemma 6.18 connects this spectral quantity to the error at the computed trial. The following theorem states the resulting convergence guarantee; its proof, including joint neighborhood invariance, is given in Section 6.4.

Theorem 2.5 (Local linear convergence of the DD-QP algorithm)**.** *Suppose Assumption 2.1 holds at an optimal pair $(X_\star, y_\star)$, and $X_\star$ is the unique primal optimum and $1 \le s := \operatorname{rank} X_\star \le r$ with $s < n$. Suppose Assumption 2.2*(ii) *holds with modulus $\mu > 0$ and radius $r_{\mathrm{QG}} > 0$. Choose $0 < \varepsilon \le \sigma_\star/(4\sqrt{r})$ and $2\rho L_{\mathrm{sp}} < 1$. Then there exist $\delta_y, \delta_X > 0$ such that, whenever*

$$\|y_0 - y_\star\| \le \delta_y, \quad X_0 \in \mathcal{D}, \quad \|X_0 - X_\star\|_F \le \delta_X,$$

Algorithm 2.2, with exact spectral computations and QP solves, is well defined and remains in this local neighborhood. Moreover, setting $d_k := \operatorname{dist}(y_k, \mathcal{Y}^*)$ and $q := (1+\rho\mu)^{-1/2} < 1$, the following hold for every $k \ge 0$:

(i) *The DD-QP solution coincides with the solution of the full s-dimensional active-space semidefinite correction; in particular,* $\operatorname{rank} X_{k+1} = s$.

(ii) *The dual distance contracts linearly:*

$$d_{k+1} \le q d_k. \tag{2.48}$$

(iii) *There exists $y_\infty \in \mathcal{Y}^*$ such that*

$$\|y_k - y_\infty\| \le \frac{2d_0}{1-q} q^k. \tag{2.49}$$

(iv) *For some constant $L_X > 0$ independent of k,*

$$\|X_{k+1} - X_\star\|_F \le L_X d_0 q^k, \qquad \|\mathcal{A}X_{k+1} - b\| \le \frac{2d_0}{\rho} q^k, \qquad F(y_{k+1}) - f^* \le \frac{d_0^2}{2\rho} q^{2k}. \tag{2.50}$$

Remark 2.6 (Local equivalence to the active spectral model)**.** The principal local property is not simply that the DD model has a small approximation error. Rather, its optimal correction matrix is identical to that of the full active-space semidefinite problem. To make the comparison precise, let $V_{k,s}$ consist of the first s columns of V_k, and define

$$\mathcal{D}_s(y_k) = \left\{ V_{k,s} W V_{k,s}^\top : W \succeq 0,\ \operatorname{tr} W \le R \right\}.$$

Then, throughout the local regime,

$$X_{k+1} = \operatorname*{argmin}_{X \in \mathcal{D}_s(y_k)} \mathcal{J}_{y_k}(X).$$

Hence the dual update is the proximal step for the full active spectral model $m_k^{\mathrm{act}}(y) = b^\top y + R[\lambda_{\max}(V_{k,s}^\top S(y) V_{k,s})]_+$. Theorem 6.12 establishes this equality of optimal corrections even for $r > s$. The comparison is with the complete s-dimensional *active* problem, not the complete r-dimensional working problem, and does not assert equality of the two feasible domains.

3 A hybrid algorithm for multi-cone problems

The single-block construction separates spectral approximation from within-subspace correction. We now apply the same distinction to heterogeneous conic problems. Blocks with inexpensive projections will be represented exactly, while large PSD blocks will use the models of Section 2. The models can be constructed blockwise, but their coefficients must in general be optimized jointly because the affine constraints couple the blocks.

3.1 Multi-cone formulation and blockwise models

Write $X = ((x_i)_{i\in\mathcal{E}}, (X_j)_{j\in\mathcal{L}})$ for the product primal variable. We consider the conic optimization problem

$$\begin{aligned} \max_X \quad & \langle C, X\rangle := \sum_{i\in\mathcal{E}}\langle C_i, x_i\rangle + \sum_{j\in\mathcal{L}}\langle C_j, X_j\rangle \\ \text{s.t.} \quad & \mathcal{A}X := \sum_{i\in\mathcal{E}}\mathcal{A}_i x_i + \sum_{j\in\mathcal{L}}\mathcal{A}_j X_j = b, \\ & x_i \in K_i, \quad i\in\mathcal{E}, \qquad X_j \succeq 0, \quad j\in\mathcal{L}. \end{aligned} \tag{3.1}$$

Here $\mathcal{E}$ indexes closed convex cones K_i with inexpensive projections, such as nonnegative orthants, second-order cones, and sufficiently small PSD cones. The set $\mathcal{L}$ indexes the large PSD blocks to be treated by spectral models. This partition is a computational choice: it determines which blocks are approximated, not which affine constraints are retained.

As in the single-block setting, assume finite, attained primal and dual optimal values with common value f^*. For a primal optimum X^*, choose $R_i > \|x_i^*\|$ for $i\in\mathcal{E}$ and $R_j > \operatorname{tr} X_j^*$ for $j\in\mathcal{L}$, and set

$$\mathcal{D}_i = K_i \cap \{x : \|x\| \le R_i\}, \qquad \mathcal{D}_j = \{X \succeq 0 : \operatorname{tr} X \le R_j\}, \qquad \mathcal{D} = \prod_\ell \mathcal{D}_\ell. \tag{3.2}$$

For each block, let $S_\ell(y) = C_\ell - \mathcal{A}_\ell^* y$. The dual cone condition is $S_\ell(y)\in K_\ell^\circ$, where K_ℓ° is the polar cone and $K_j = \mathbb{S}_+^{n_j}$ for $j\in\mathcal{L}$. The exact penalty associated with (3.1) is

$$F(y) = b^\top y + \sum_{i\in\mathcal{E}} R_i\|\Pi_{K_i}(S_i(y))\| + \sum_{j\in\mathcal{L}} R_j[\lambda_{\max}(S_j(y))]_+. \tag{3.3}$$

The strict radii give the same exact-penalty property as before: $\min F = f^*$, and $\mathcal{Y}^* := \operatorname{argmin} F$ is the dual optimal set. The compact product domain will also allow the single-block support-model estimates to be used without changing their form.

For any product model $\mathcal{K} = \prod_\ell \mathcal{K}_\ell \subseteq \mathcal{D}$, define the common quadratic correction objective

$$\mathcal{J}_y(X) := \frac{\rho}{2}\|\mathcal{A}X - b\|^2 - \sum_\ell \langle S_\ell(y), X_\ell\rangle. \tag{3.4}$$

Given an exact correction

$$\widehat{X} \in \operatorname*{argmin}_{X\in\mathcal{K}} \mathcal{J}_y(X), \tag{3.5}$$

the associated dual trial point is

$$\widehat{y} = y + \rho(\mathcal{A}\widehat{X} - b). \tag{3.6}$$

The only coupling in this objective is through $\mathcal{A}X - b$. This permits blockwise model construction, but not independent minimization of the block corrections. We next specify the product models used in the two phases.

Diagonal product model. At accepted center y_k, let $X_{k,0} = X_k$ and $y_{k,0} = y_k$. At inner iteration i, for each $j\in\mathcal{L}$, compute

$$V_{j,k,i} = \operatorname{TopEig}_{r_j^{\mathrm{d}}}\big(S_j(y_{k,i})\big),$$

and define

$$\mathcal{K}_{j,k,i}^{\mathrm{d}} = \left\{\beta_j X_{j,k,i} + R_j V_{j,k,i}\operatorname{Diag}(w_j)V_{j,k,i}^\top : \beta_j \ge 0,\ w_j \ge 0,\ \beta_j + \mathbf{1}^\top w_j \le 1\right\}. \tag{3.7}$$

The complete diagonal model is

$$\mathcal{K}_{k,i}^{\mathrm{d}} = \prod_{\ell\in\mathcal{E}} \mathcal{D}_\ell \times \prod_{j\in\mathcal{L}} \mathcal{K}_{j,k,i}^{\mathrm{d}}. \tag{3.8}$$

Its joint correction and trial point are

$$X_{k,i+1} \in \operatorname*{argmin}_{X \in \mathcal{K}^{\mathrm{d}}_{k,i}} \mathcal{J}_{y_k}(X), \qquad y_{k,i+1} = y_k + \rho(\mathcal{A}X_{k,i+1} - b). \tag{3.9}$$

As in the single-block algorithm, the center y_k is fixed during the entire refinement loop; only the product model changes.

Let

$$m^{\mathrm{d}}_{k,i}(y) = b^\top y + \sum_{\ell \in \mathcal{E}} \sigma_{\mathcal{D}_\ell}(S_\ell(y)) + \sum_{j \in \mathcal{L}} \sigma_{\mathcal{K}^{\mathrm{d}}_{j,k,i}}(S_j(y)).$$

The predicted reduction and model error are

$$\Delta_{k,i} = F(y_k) - m^{\mathrm{d}}_{k,i}(y_{k,i+1}), \qquad e_{k,i} = F(y_{k,i+1}) - m^{\mathrm{d}}_{k,i}(y_{k,i+1}). \tag{3.10}$$

We accept the trial whenever

$$e_{k,i} \le \frac{\Delta_{k,i}}{2} \qquad \text{or} \qquad e_{k,i} \le \eta_{\mathrm{pre}}. \tag{3.11}$$

On rejection, the center is retained. Each large block keeps its new primal output as an aggregate and uses the spectrum at the common trial multiplier to generate the next model. The exact blocks retain their complete sets throughout.

DD product model. In the local phase, the large blocks instead use primal-adapted DD coordinates. For the current center y_k and accepted primal blocks $X_{j,k}$, compute

$$V_{j,k} \in \mathrm{TopEig}_{r_j}\big(S_j(y_k)\big), \qquad L_{j,k} = \mathrm{chol}\left(V_{j,k}^\top X_{j,k} V_{j,k} + \varepsilon_j I\right), \tag{3.12}$$

where $\mathrm{chol}(Z)$ is the upper triangular matrix with positive diagonal satisfying $Z = \mathrm{chol}(Z)\,\mathrm{chol}(Z)^\top$. Define

$$\mathcal{K}_{j,k} = \left\{ V_{j,k} L_{j,k} T_j L_{j,k}^\top V_{j,k}^\top : T_j \in \mathsf{DD}_{r_j},\ \mathrm{tr}(L_{j,k} T_j L_{j,k}^\top) \le R_j \right\}. \tag{3.13}$$

The joint DD model is

$$\mathcal{K}^{\mathrm{DD}}_k = \prod_{\ell \in \mathcal{E}} \mathcal{D}_\ell \times \prod_{j \in \mathcal{L}} \mathcal{K}_{j,k}. \tag{3.14}$$

The DD phase performs the direct joint correction

$$X_{k+1} \in \operatorname*{argmin}_{X \in \mathcal{K}^{\mathrm{DD}}_k} \mathcal{J}_{y_k}(X), \qquad y_{k+1} = y_k + \rho(\mathcal{A}X_{k+1} - b). \tag{3.15}$$

Unlike the diagonal phase, this phase uses one joint correction per center update and has no inner acceptance loop.

Numerical solution of the joint correction. Using the coefficient representations of Models 2 and 3, both product corrections become convex quadratic conic subproblems in the retained variables x_i and the spectral coefficients. Their feasible sets in these variables are Cartesian products of the sections $\mathcal{D}_i$ and simplex-type coefficient sets. Projection onto $\mathcal{D}_i$ is obtained by projecting onto K_i and, if necessary, rescaling to radius R_i; the coefficient sets admit simplex-type projections. Thus the joint subproblems can be solved numerically by projected-gradient methods, such as a safeguarded projected Barzilai–Borwein method. The gradient retains the coupling through $\mathcal{A}X - b$, even though these projections are blockwise. The convergence analysis assumes exact minimization of each joint subproblem and counts one such minimization as one model solve; it does not count the numerical iterations used to obtain that correction.

3.2 The hybrid diagonal-to-DD algorithm

The hybrid begins with the diagonal algorithm and switches permanently to the direct DD iteration. Let Δ_k^{acc} be the predicted reduction of the accepted diagonal trial at iteration k. The practical switching rule is

$$\Delta_k^{\mathrm{acc}} \le \tau_{\mathrm{sw}} \qquad \text{or} \qquad k+1 \ge K_{\mathrm{pre}}, \tag{3.16}$$

where $\tau_{\mathrm{sw}} > 0$ and $K_{\mathrm{pre}} < \infty$ are user-specified warm-start parameters. The first condition detects small predicted progress, and the second bounds the duration of the warm start. Neither condition certifies that the primal matrices and dual center lie in the neighborhood required by the DD theorem. Accordingly, the local guarantee below is conditional on the switching state. Without that condition, the unsafeguarded DD phase is not covered by a global convergence result.

Algorithm 3.1 Hybrid multi-cone algorithm: diagonal warm start and DD local phase

Input: $X_0 \in \mathcal{D}$, $y_0 \in \mathbb{R}^m$, $\rho > 0$, diagonal ranks $\{r_j^{\mathrm{d}}\}$, DD ranks $\{r_j\}$, regularizations $\varepsilon_j > 0$, all block radii $\{R_\ell\}$, and warm-start parameters $\eta_{\mathrm{pre}}, \tau_{\mathrm{sw}} > 0$ and $1 \le K_{\mathrm{pre}} < \infty$.

Set phase = diag.
for $k = 0, 1, 2, \ldots$ **do**
  if phase = diag **then**
    Initialize $(X_{k,0}, y_{k,0}) = (X_k, y_k)$.
    for $i = 0, 1, 2, \ldots$ **do**
      Construct the joint diagonal model $\mathcal{K}_{k,i}^{\mathrm{d}}$.
      Solve the joint correction (3.9).
      Compute $(\Delta_{k,i}, e_{k,i})$.
      if (3.11) holds **then**
        Accept $(X_{k+1}, y_{k+1}) = (X_{k,i+1}, y_{k,i+1})$ and set $\Delta_k^{\mathrm{acc}} = \Delta_{k,i}$.
        break
      end if
    end for
    if (3.16) holds **then**
      Set phase = DD.
    end if
  else
    Construct the joint DD model $\mathcal{K}_k^{\mathrm{DD}}$.
    Solve the joint DD correction (3.15).
  end if
end for

The large-block coefficient dimension changes from $O(r_j^{\mathrm{d}})$ in the diagonal phase to $O(r_j^2)$ in the DD phase. The former pays for model refinement, whereas the latter uses the richer model in one direct correction. In both phases, the exact cone sets are unchanged and all blocks are optimized against the same residual $\mathcal{A}X - b$. Thus the switch changes the approximation of the large PSD blocks, not the coupling between them.

Convergence guarantees. The key connection to the common analysis is an additive decomposition of model error. For $m(y) = b^\top y + \sigma_{\mathcal{K}}(S(y))$, we have

$$F(y) - m(y) = \sum_{j \in \mathcal{L}} \left[\sigma_{\mathcal{D}_j}(S_j(y)) - \sigma_{\mathcal{K}_j}(S_j(y))\right], \tag{3.17}$$

since $\mathcal{K}_i = \mathcal{D}_i$ for every $i \in \mathcal{E}$. The exact blocks therefore introduce no approximation error, while the large-block spectral errors add at the same trial multiplier. This identity permits both phases to use the common analysis in Section 4. For the improved diagonal bound, (2.32) is understood for the complete penalty (3.3) on a relevant sublevel containing y_0. The local DD guarantee uses the following additional geometry.

Assumption 3.1. There is an optimal pair $(X_\star, y_\star)$ with $X_\star \in \mathcal{D}$ satisfying the following conditions.

(i) The large PSD blocks are uniquely determined among primal optimal solutions in $\mathcal{D}$: every such solution $X = ((x_i)_{i\in\mathcal{E}}, (X_j)_{j\in\mathcal{L}})$ satisfies $X_j = X_{j,\star}$ for all $j \in \mathcal{L}$.

(ii) For every $j \in \mathcal{L}$, let $Z_{j,\star} := -S_j(y_\star)$ and $s_j := \operatorname{rank} X_{j,\star}$. Then

$$1 \le s_j < n_j, \qquad X_{j,\star}Z_{j,\star} = 0, \qquad X_{j,\star} + Z_{j,\star} \succ 0.$$

(iii) There exist $\mu, \delta > 0$ such that the complete penalty satisfies

$$F(y) - f^* \ge \frac{\mu}{2}\operatorname{dist}^2(y, \mathcal{Y}^*) \qquad \text{whenever } \|y - y_\star\| \le \delta.$$

For conic programs over nonnegative orthants, second-order cones, and positive semidefinite cones, the quadratic growth condition can be implied by strict complementarity over all cone blocks, according to the error-bound framework developed by Ding and Udell [50, Sections 2–3 and Remark 4].

For the local DD phase, let $\sigma_j := \lambda_{s_j}(X_{j,\star}) > 0$ and let $\gamma_j > 0$ be the smallest eigenvalue of $Z_{j,\star}$ on $(\operatorname{range} X_{j,\star})^\perp$. Define

$$L_{\mathrm{mix}} := 2\sum_{j\in\mathcal{L}} \frac{R_j\|\mathcal{A}_j\|^2}{\gamma_j}, \tag{3.18}$$

and choose

$$s_j \le r_j \le n_j, \qquad 0 < \varepsilon_j \le \frac{\sigma_j}{4\sqrt{r_j}} \quad (j \in \mathcal{L}), \qquad 2\rho L_{\mathrm{mix}} < 1.$$

Theorem 3.1 (Multi-cone convergence guarantees)**.** *Under the standing solution and strict-radius assumptions for (3.1), consider Algorithm 3.1 with exact joint corrections and spectral oracles. Count every joint correction, including a rejected trial, as one model solve.*

(i) Diagonal phase. *With switching disabled and $\eta_{\mathrm{pre}} = \eta > 0$, every refinement loop terminates finitely. For the accepted averages $\bar{X}_K = K^{-1}\sum_{k=1}^K X_k$ and $\bar{y}_K = K^{-1}\sum_{k=1}^K y_k$, there is a constant $\Gamma > 0$, independent of K and η, such that*

$$\max\left\{F(\bar{y}_K) - f^*,\ \|\mathcal{A}\bar{X}_K - b\|,\ |\langle C, \bar{X}_K\rangle - f^*|\right\} \le \frac{\Gamma}{K} + 2\eta. \tag{3.19}$$

Taking $\eta = \epsilon/4$ attains all three errors at most ϵ in $O(\epsilon^{-2})$ model solves. Under the relevant-sublevel quadratic-growth condition above, a suitable fixed $\eta = \Theta(\epsilon)$ attains a current-center penalty gap at most ϵ in $O(\epsilon^{-1})$ model solves.

(ii) Local DD phase. *Under Assumption 3.1 and the local parameter choices above, there is an invariant joint neighborhood of $y_\star$ and $\{X_{j,\star}\}_{j\in\mathcal{L}}$. If the switching state at k_{sw} lies in this neighborhood, every subsequent DD correction has the same minimizer set as the full active-space product correction and induces a unique dual update satisfying*

$$\operatorname{dist}(y_{k+1}, \mathcal{Y}^*) \le (1 + \rho\mu)^{-1/2}\operatorname{dist}(y_k, \mathcal{Y}^*) \qquad (k \ge k_{\mathrm{sw}}). \tag{3.20}$$

The dual iterates converge R-linearly to an optimum, and the equality residuals converge R-linearly to zero. Every new large-block output $X_{j,k+1}$ has rank s_j and converges to $X_{j,\star}$. Dual distance ϵ is attained in $N_{\mathrm{pre}} + O(\log(1/\epsilon))$ total model solves, where N_{pre} counts all corrections before the switch.

4 A unified framework for convergence analysis

We now establish the estimates that underlie the diagonal and DD guarantees. The common question is how accurately a reduced support model must approximate the exact penalty in order for its correction to make progress. This question can be studied on a compact convex primal domain, without using any particular spectral representation.

The analysis proceeds in three stages. We first relate primal correction optimality to a dual proximal step and derive a primal–dual comparison inequality. We then show how the mixed relative–absolute

accuracy test gives averaged guarantees, and how quadratic growth strengthens these to current-iterate bounds. Finally, we replace the fixed error allowance by quadratic model accuracy and obtain a linear convergence result. Throughout this section, k counts completed center updates. The cost of obtaining the required accuracy is not assumed away: it is analyzed for the diagonal model in Section 5 and for the DD model in Section 6.

4.1 Common correction and basic estimates

Let $\mathcal{D}$ be a nonempty compact convex set in a finite-dimensional Euclidean space. Consider the primal problem $\max\{\langle C, X\rangle : \mathcal{A}X = b,\ X \in \mathcal{D}\}$, and define

$$\mathscr{L}(X, y) := \langle C, X\rangle + \langle y, b - \mathcal{A}X\rangle, \qquad F(y) := \max_{X\in\mathcal{D}} \mathscr{L}(X, y). \tag{4.1}$$

We assume that the primal optimal value and $\min F$ are finite, attained, and equal to f^*. Write $\mathcal{Y}^* := \operatorname{argmin} F$. The function F is finite and convex, and $\mathcal{Y}^*$ is nonempty, closed, and convex.

For a nonempty compact set $\mathcal{K}$, its support function is $\sigma_{\mathcal{K}}(H) := \max_{X\in\mathcal{K}}\langle H, X\rangle$. At center y_k, let $\mathcal{K}_k \subseteq \mathcal{D}$ be a nonempty compact convex model, and set

$$m_k(y) := \max_{X\in\mathcal{K}_k} \mathscr{L}(X, y) = b^\top y + \sigma_{\mathcal{K}_k}(C - \mathcal{A}^* y) \le F(y). \tag{4.2}$$

For a fixed $\rho > 0$, the exact correction and its dual update are

$$\begin{aligned} &X_{k+1} \in \operatorname*{argmin}_{X\in\mathcal{K}_k} \mathcal{J}_{y_k}(X), \qquad \mathcal{J}_y(X) := \frac{\rho}{2}\|\mathcal{A}X - b\|^2 - \langle C - \mathcal{A}^* y, X\rangle, \\ &y_{k+1} = y_k + \rho(\mathcal{A}X_{k+1} - b). \end{aligned} \tag{4.3}$$

Compactness guarantees existence of a correction. The model may depend on previous outputs; it need not be nested with earlier models or interpolate F at the center. We index its primal output by the completed dual update: X_{k+1} is associated with $y_k \mapsto y_{k+1}$.

Write $s_k := y_{k+1} - y_k$ and define

$$\begin{aligned} e_k &:= F(y_{k+1}) - m_k(y_{k+1}), \quad \Delta_k := F(y_k) - m_k(y_{k+1}), \\ h_k &:= F(y_k) - f^*, \qquad\qquad\ \ d_k := \operatorname{dist}(y_k, \mathcal{Y}^*). \end{aligned} \tag{4.4}$$

Thus e_k measures the error at the computed update, whereas Δ_k is the predicted decrease before subtracting the proximal quadratic. Their difference is the actual decrease, $h_k - h_{k+1} = \Delta_k - e_k$.

The following lemma connects the primal correction with the dual support model and provides the comparison inequality used in our convergence analysis. The support and proximal identities are standard in spectral bundle methods; see [27]. We include a short proof for general compact convex models.

Lemma 4.1 (Support corrections and proximal comparison)**.** *Every exact correction* (4.3) *satisfies*

$$m_k(y_{k+1}) = \mathscr{L}(X_{k+1}, y_{k+1}), \qquad b - \mathcal{A}X_{k+1} = -s_k/\rho \in \partial m_k(y_{k+1}). \tag{4.5}$$

Its dual update is the unique proximal minimizer

$$y_{k+1} = \operatorname*{argmin}_{y\in\mathbb{R}^m} \left\{ m_k(y) + \frac{\|y - y_k\|^2}{2\rho} \right\}. \tag{4.6}$$

Moreover,

$$e_k \ge 0, \qquad \Delta_k \ge \frac{\|s_k\|^2}{\rho}, \qquad h_k - h_{k+1} = \Delta_k - e_k. \tag{4.7}$$

For every $X \in \mathcal{D}$ and $y \in \mathbb{R}^m$,

$$\mathscr{L}(X, y_{k+1}) - \mathscr{L}(X_{k+1}, y) \le e_k + \frac{\|y_k - y\|^2 - \|y_{k+1} - y\|^2 - \|s_k\|^2}{2\rho}. \tag{4.8}$$

Consequently,

$$F(y_{k+1}) - F(y) \le e_k + \frac{\|y_k - y\|^2 - \|y_{k+1} - y\|^2 - \|s_k\|^2}{2\rho}. \tag{4.9}$$

Proof. Since $\nabla \mathcal{J}_{y_k}(X_{k+1}) = -(C - \mathcal{A}^* y_{k+1})$, first-order optimality implies that X_{k+1} maximizes $\mathscr{L}(\cdot, y_{k+1})$ over $\mathcal{K}_k$. This proves the support identity. For every $y \in \mathbb{R}^m$, we then have

$$m_k(y) \ge \mathscr{L}(X_{k+1}, y) = m_k(y_{k+1}) - \frac{\langle s_k, y - y_{k+1} \rangle}{\rho}.$$

Thus $-s_k/\rho \in \partial m_k(y_{k+1})$, which is precisely the optimality condition for the strongly convex problem (4.6). Taking $y = y_k$ in the above inequality and using $m_k \le F$ gives $\Delta_k \ge \|s_k\|^2/\rho$. The remaining relations in (4.7) follow from $m_k \le F$ and the definitions.

Next, for $X \in \mathcal{D}$ and $y \in \mathbb{R}^m$, split

$$\begin{aligned}\mathscr{L}(X, y_{k+1}) - \mathscr{L}(X_{k+1}, y) = \mathscr{L}(X, y_{k+1}) - \mathscr{L}(X_{k+1}, y_{k+1}) \\ + \mathscr{L}(X_{k+1}, y_{k+1}) - \mathscr{L}(X_{k+1}, y).\end{aligned}$$

The support identity bounds the first difference by

$$\mathscr{L}(X, y_{k+1}) - \mathscr{L}(X_{k+1}, y_{k+1}) \le F(y_{k+1}) - m_k(y_{k+1}) = e_k.$$

For the second difference, affine dependence on the dual argument and $b - \mathcal{A}X_{k+1} = -s_k/\rho$ give

$$\begin{aligned}\mathscr{L}(X_{k+1}, y_{k+1}) - \mathscr{L}(X_{k+1}, y) &= \langle y_{k+1} - y, b - \mathcal{A}X_{k+1} \rangle \\ &= -\frac{\langle s_k, y_{k+1} - y \rangle}{\rho}.\end{aligned}$$

Adding these estimates yields

$$\mathscr{L}(X, y_{k+1}) - \mathscr{L}(X_{k+1}, y) \le e_k - \frac{\langle s_k, y_{k+1} - y \rangle}{\rho}.$$

Expanding $\|y_k - y\|^2 = \|y_{k+1} - y - s_k\|^2$ yields (4.8). Maximizing over $X \in \mathcal{D}$ and using $\mathscr{L}(X_{k+1}, y) \le F(y)$ then proves (4.9).

□

4.2 Mixed accuracy and averaged convergence

We begin with the accuracy condition used by the diagonal algorithm:

$$e_k \le \Delta_k/2 \qquad \text{or} \qquad e_k \le \eta, \qquad \eta \ge 0, \tag{4.10}$$

where the tolerance is fixed throughout a run. A relative acceptance ensures objective decrease, while an absolute-only acceptance can increase F by at most η. Consequently, objective monotonicity alone is not the appropriate quantity to track. We instead combine the dual objective gap with the squared distance to the optimal set:

$$E_k := d_k^2 + 2\rho h_k, \qquad C_0 := \frac{E_0}{2\rho} = h_0 + \frac{d_0^2}{2\rho}. \tag{4.11}$$

Lemma 4.2 (Mixed accuracy and the common energy)**.** *Under* (4.10)*,*

$$e_k \le h_k - h_{k+1} + 2\eta, \qquad h_{k+1} \le h_k + \eta, \tag{4.12}$$

$$\frac{\|s_k\|^2}{\rho} \le \Delta_k \le 2(h_k - h_{k+1}) + 2\eta, \tag{4.13}$$

and

$$E_{k+1} \le E_k - \|s_k\|^2 - 2\rho h_{k+1} + 4\rho\eta. \tag{4.14}$$

The same energy estimate holds with E_k replaced by $\|y_k - y\|^2 + 2\rho h_k$ for any fixed $y \in \mathcal{Y}^*$. On a relative step, the additive term $4\rho\eta$ can be omitted.

Proof. First consider the two branches of the test. On a relative step, $e_k \le \Delta_k - e_k = h_k - h_{k+1}$ and $h_{k+1} \le h_k$. On an absolute step, $e_k \le \eta$ and $h_k - h_{k+1} = \Delta_k - e_k \ge -\eta$. Hence $e_k \le h_k - h_{k+1} + 2\eta$ and $h_{k+1} \le h_k + \eta$. This establishes (4.12) in both cases. Substituting its first bound into $\Delta_k = h_k - h_{k+1} + e_k$ and invoking (4.7) yields (4.13).

For any fixed $y \in \mathcal{Y}^*$, (4.9) gives

$$\begin{aligned}\|y_{k+1} - y\|^2 &\le \|y_k - y\|^2 - \|s_k\|^2 - 2\rho h_{k+1} + 2\rho e_k \\ &\le \|y_k - y\|^2 - \|s_k\|^2 + 2\rho h_k - 4\rho h_{k+1} + 4\rho\eta.\end{aligned}$$

Adding $2\rho h_{k+1}$ proves the fixed-reference energy inequality. Choosing y nearest to y_k and using $d_{k+1}^2 \le \|y_{k+1} - y\|^2$ then gives (4.14). For a relative step, the sharper bound $e_k \le h_k - h_{k+1}$ removes the additive error term. This completes the proof. □

Lemma 4.2 controls the cumulative model error and the dual energy. To obtain primal guarantees as well, we return to the saddle comparison in Lemma 4.1. Averaging makes its distance terms telescope, and a bounded set of comparison multipliers allows us to extract both feasibility and objective bounds. The resulting averages are computable without an optimal solution.

Theorem 4.3 (Averaged primal and dual guarantees)**.** *Suppose the exact corrections* (4.3) *satisfy* (4.10) *with fixed* $\eta \ge 0$*. Choose* $y^{\mathrm{ref}} \in \mathcal{Y}^*$ *and define*

$$\bar{X}_K := \frac{1}{K}\sum_{k=0}^{K-1} X_{k+1}, \qquad \bar{y}_K := \frac{1}{K}\sum_{k=0}^{K-1} y_{k+1}, \qquad \Gamma_0 := h_0 + \frac{(\|y_0\| + 2\|y^{\mathrm{ref}}\| + 1)^2}{2\rho}. \tag{4.15}$$

Then $\bar{X}_K \in \mathcal{D}$ *and, for every* $K \ge 1$*,*

$$\max\left\{F(\bar{y}_K) - f^*,\ \|\mathcal{A}\bar{X}_K - b\|, |\langle C, \bar{X}_K\rangle - f^*|\right\} \le \frac{\Gamma_0}{K} + 2\eta. \tag{4.16}$$

Moreover,

$$\mathcal{A}\bar{X}_K - b = \frac{y_K - y_0}{\rho K}. \tag{4.17}$$

Proof. We first derive a common averaged saddle inequality, then extract the three errors in the theorem. Convexity gives $\bar{X}_K \in \mathcal{D}$, and summing the dual updates yields (4.17). Moreover, (4.12) implies

$$\sum_{k=0}^{K-1} e_k \le h_0 - h_K + 2K\eta \le h_0 + 2K\eta.$$

For fixed $X \in \mathcal{D}$ and $y \in \mathbb{R}^m$, sum the primal–dual comparison (4.8) and divide by K. The distance terms telescope. Since $\mathscr{L}$ is affine in each argument, dropping the nonpositive terminal-distance and squared-step terms gives

$$\mathscr{L}(X, \bar{y}_K) - \mathscr{L}(\bar{X}_K, y) \le \frac{h_0 + \|y_0 - y\|^2/(2\rho)}{K} + 2\eta. \tag{4.18}$$

Set $B_\star := 2\|y^{\mathrm{ref}}\| + 1$, $r_K := \mathcal{A}\bar{X}_K - b$, and $\epsilon_K := \Gamma_0/K + 2\eta$. For every $\|y\| \le B_\star$, $\|y_0 - y\| \le \|y_0\| + B_\star$, so the right-hand side of (4.18) is at most ϵ_K uniformly on this ball. We may therefore maximize its left-hand side over $X \in \mathcal{D}$ and $\|y\| \le B_\star$. Since $\max_{X\in\mathcal{D}} \mathscr{L}(X, \bar{y}_K) = F(\bar{y}_K)$, $\mathscr{L}(\bar{X}_K, y) = \langle C, \bar{X}_K\rangle - \langle y, r_K\rangle$, and $\max_{\|y\|\le B_\star}\langle y, r_K\rangle = B_\star\|r_K\|$, we obtain

$$\mathcal{G}_K := F(\bar{y}_K) - \langle C, \bar{X}_K\rangle + B_\star\|r_K\| \le \epsilon_K. \tag{4.19}$$

Because $\bar{X}_K \in \mathcal{D}$ and y^{ref} is optimal, $\mathscr{L}(\bar{X}_K, y^{\mathrm{ref}}) \le F(y^{\mathrm{ref}}) = f^*$. Expanding this inequality gives

$$\langle C, \bar{X}_K\rangle - f^* \le \langle y^{\mathrm{ref}}, r_K\rangle \le \|y^{\mathrm{ref}}\|\|r_K\|.$$

Consequently,

$$\mathcal{G}_K = F(\bar{y}_K) - f^* - (\langle C, \bar{X}_K\rangle - f^*) + B_\star\|r_K\| \geq F(\bar{y}_K) - f^* + (\|y^{\mathrm{ref}}\| + 1)\|r_K\|.$$

Both terms are nonnegative, and hence

$$0 \leq F(\bar{y}_K) - f^* \leq \epsilon_K, \qquad \|r_K\| \leq \frac{\epsilon_K}{\|y^{\mathrm{ref}}\| + 1} \leq \epsilon_K.$$

For the positive part of the primal objective error,

$$[\langle C, \bar{X}_K\rangle - f^*]_+ \leq \|y^{\mathrm{ref}}\|\|r_K\| \leq \frac{\|y^{\mathrm{ref}}\|}{\|y^{\mathrm{ref}}\| + 1}\epsilon_K \leq \epsilon_K.$$

For the negative part, rearranging (4.19) gives

$$f^* - \langle C, \bar{X}_K\rangle = \mathcal{G}_K - (F(\bar{y}_K) - f^*) - B_\star\|r_K\| \leq \mathcal{G}_K \leq \epsilon_K.$$

The two one-sided estimates prove the absolute primal objective bound. Together with the dual and residual estimates, they establish (4.16).

□

Theorem 4.3 gives $O(K^{-1}) + O(\eta)$ error bounds for a fixed positive tolerance. The error floor is not removed unless the accuracy regime is changed. Averaging preserves membership in $\mathcal{D}$, although it may increase the rank of a matrix output. Also, this theorem analyzes a sequence of completed corrections; it does not assert that a particular refinement procedure produces each correction in finite time. That property must be verified for the concrete algorithm.

4.3 Quadratic growth and current-iterate convergence

The averaged guarantee does not yet control the current iterate at a geometric rate. We next show that quadratic growth provides the additional link. As in Section 2.4, we restrict attention to the multiplier directions visited by the method:

$$\mathcal{E}_0 := y_0 + \operatorname{range}\mathcal{A}, \qquad \mathcal{L}_H := \{y : F(y) - f^* \leq H\}, \qquad H > 0. \tag{4.20}$$

Primal feasibility gives $b \in \operatorname{range}\mathcal{A}$, so all dual updates remain in $\mathcal{E}_0$.

Assume that, for some $\mu > 0$, the complete penalty satisfies

$$F(y) - f^* \geq \frac{\mu}{2}\operatorname{dist}(y, \mathcal{Y}^*)^2 \qquad (y \in \mathcal{E}_0 \cap \mathcal{L}_H). \tag{4.21}$$

Before applying this bound at the next iterate, we must verify that the algorithm stays in its validity region. The next lemma does so by first controlling predicted decrease using growth only at the current center. This order avoids assuming the desired invariance.

Lemma 4.4 (Predicted decrease and sublevel invariance)**.** *At any center satisfying $h_k \geq \mu d_k^2/2$, an exact correction obeys*

$$\Delta_k \geq c_{\mathrm{QG}} h_k, \qquad c_{\mathrm{QG}} := \min\{1/2, \rho\mu/4\}. \tag{4.22}$$

This conclusion does not require the mixed accuracy test. If (4.21) holds, $h_0 \leq H$, and the completed updates satisfy (4.10) with $0 \leq \eta \leq c_{\mathrm{QG}}H/2$, then all centers belong to $\mathcal{E}_0 \cap \mathcal{L}_H$. Moreover, $\operatorname{proj}_{\mathcal{Y}^*}(y_0) \in \mathcal{E}_0 \cap \mathcal{L}_H$, *and*

$$h_{k+1} \leq (1 - c_{\mathrm{QG}}/2)h_k + \eta. \tag{4.23}$$

Proof. To bound the predicted decrease, compare the proximal correction with points on the segment from the current center to a nearest optimum. Let y_k^{ref} be nearest to y_k and set $y(t) = (1-t)y_k + ty_k^{\mathrm{ref}}$ for $0 \le t \le 1$. Proximal optimality, the minorant relation $m_k \le F$, and convexity yield

$$m_k(y_{k+1}) + \frac{\|s_k\|^2}{2\rho} \le m_k(y(t)) + \frac{\|y(t) - y_k\|^2}{2\rho} \le (1-t)F(y_k) + tf^* + \frac{t^2 d_k^2}{2\rho}.$$

Rearranging and maximizing over t yields

$$\Delta_k - \frac{\|s_k\|^2}{2\rho} \ge \max_{0 \le t \le 1} \left\{ th_k - \frac{t^2 d_k^2}{2\rho} \right\}. \tag{4.24}$$

If $h_k \ge \mu d_k^2/2$, the right-hand side is at least $h_k \max_{0\le t\le 1}\{t - t^2/(\rho\mu)\}$. For $\rho\mu \le 2$, choosing $t = \rho\mu/2$ gives the coefficient $\rho\mu/4$. For $\rho\mu > 2$, choosing $t = 1$ gives $1 - 1/(\rho\mu) > 1/2$. Dropping the nonnegative squared-step term proves (4.22).

Now suppose $y_k \in \mathcal{E}_0 \cap \mathcal{L}_H$. A relative acceptance satisfies

$$h_{k+1} = h_k - \Delta_k + e_k \le h_k - \Delta_k/2 \le (1 - c_{\mathrm{QG}}/2)h_k.$$

An absolute-only acceptance instead satisfies

$$h_{k+1} \le h_k - \Delta_k + \eta \le (1 - c_{\mathrm{QG}})h_k + \eta \le (1 - c_{\mathrm{QG}}/2)h_k + \eta.$$

Both branches therefore satisfy (4.23). If $h_k \le H$ and $\eta \le c_{\mathrm{QG}}H/2$, its right-hand side is at most H. The update also stays in $\mathcal{E}_0$. Starting from $h_0 \le H$, induction proves that every accepted center remains in the region where quadratic growth is available.

It remains to locate the nearest optimum to y_0. The adjoint relation gives the orthogonal decomposition

$$\mathbb{R}^m = \operatorname{range}\mathcal{A} \oplus \ker\mathcal{A}^*.$$

Write $y_0 = y_{0,\|} + y_{0,\perp}$ according to this decomposition. For $u \in \ker\mathcal{A}^*$, primal feasibility implies $b^\top u = \langle X^*, \mathcal{A}^* u\rangle = 0$, and hence $F(y+u) = F(y)$. Let $y^{\mathrm{ref}} = \operatorname{proj}_{\mathcal{Y}^*}(y_0)$, which exists because $\mathcal{Y}^*$ is nonempty, closed, and convex, and write $y^{\mathrm{ref}} = y_{\|}^{\mathrm{ref}} + y_{\perp}^{\mathrm{ref}}$. Replacing its perpendicular component by $y_{0,\perp}$ preserves optimality. By orthogonality,

$$\|y^{\mathrm{ref}} - y_0\|^2 = \|y_{\|}^{\mathrm{ref}} - y_{0,\|}\|^2 + \|y_{\perp}^{\mathrm{ref}} - y_{0,\perp}\|^2.$$

Minimality of this distance therefore forces $y_{\perp}^{\mathrm{ref}} = y_{0,\perp}$, so $y^{\mathrm{ref}} \in y_0 + \operatorname{range}\mathcal{A} = \mathcal{E}_0$. Since $F(y^{\mathrm{ref}}) = f^*$, it also belongs to $\mathcal{L}_H$. □

We can now apply quadratic growth along the accepted sequence. Its role is to bound the common energy by the objective gap; the energy decrease in Lemma 4.2 then becomes a geometric recurrence with an additive accuracy term. The primal residual follows from the size of the dual step. A bound on the raw primal objective requires a further multiplier estimate, which is stated separately.

Theorem 4.5 (Current primal–dual bounds under quadratic growth)**.** *Suppose* (4.21) *holds with* $\mu > 0$*,* $h_0 \le H$*, and every exact correction satisfies* (4.10) *with* $0 \le \eta \le c_{\mathrm{QG}}H/2$*. Set*

$$q_{\mathrm{mix}} := \frac{1+\rho\mu}{1+2\rho\mu} < 1, \qquad B := \frac{2(1+\rho\mu)}{\rho\mu}. \tag{4.25}$$

All accepted centers remain in $\mathcal{E}_0 \cap \mathcal{L}_H$*, and, for every* $k \ge 0$*,*

$$E_k \le q_{\mathrm{mix}}^k E_0 + 2\rho B\eta(1 - q_{\mathrm{mix}}^k). \tag{4.26}$$

In particular,

$$h_k \le C_0 q_{\mathrm{mix}}^k + B\eta, \qquad d_k^2 \le 2\rho(C_0 q_{\mathrm{mix}}^k + B\eta). \tag{4.27}$$

Let $r_k^{\mathrm{p}} := \sqrt{2\{C_0 q_{\mathrm{mix}}^k + (B+1)\eta\}/\rho}$. *The raw primal output satisfies* $\|\mathcal{A}X_{k+1} - b\| \le r_k^{\mathrm{p}}$. *If, in addition,* $M_H := \sup_{y \in \mathcal{E}_0 \cap \mathcal{L}_H} \|y\| < \infty$, *then the raw primal objective satisfies*

$$|\langle C, X_{k+1}\rangle - f^*| \le C_0 q_{\mathrm{mix}}^k + (B+2)\eta + M_H r_k^{\mathrm{p}}. \tag{4.28}$$

The boundedness condition is needed only for the raw objective bound.

Proof. Dual energy. Lemma 4.4 places the complete accepted sequence in the growth region. We may therefore use quadratic growth at the new center to obtain

$$E_{k+1} = d_{k+1}^2 + 2\rho h_{k+1} \le 2(1/\mu + \rho) h_{k+1},$$

which implies $2\rho h_{k+1} \ge \rho\mu E_{k+1}/(1+\rho\mu)$. Substitution into (4.14), followed by removal of the nonpositive squared-step term, gives

$$\left(1 + \frac{\rho\mu}{1+\rho\mu}\right) E_{k+1} \le E_k + 4\rho\eta, \qquad E_{k+1} \le q_{\mathrm{mix}} E_k + 4\rho\eta q_{\mathrm{mix}}.$$

For $k \ge 1$, iterating this recurrence yields

$$E_k \le q_{\mathrm{mix}}^k E_0 + 4\rho\eta q_{\mathrm{mix}} \sum_{j=0}^{k-1} q_{\mathrm{mix}}^j = q_{\mathrm{mix}}^k E_0 + 4\rho\eta \frac{1+\rho\mu}{\rho\mu}(1 - q_{\mathrm{mix}}^k) = q_{\mathrm{mix}}^k E_0 + 2\rho B\eta(1 - q_{\mathrm{mix}}^k).$$

The estimate also holds at $k = 0$. The inequalities $h_k \le E_k/(2\rho)$, $d_k^2 \le E_k$, and $1 - q_{\mathrm{mix}}^k \le 1$ now give (4.26)–(4.27).

Raw primal feasibility. To transfer the dual estimate to the primal output, use the identity between its equality residual and the dual step. Together with (4.13), it gives

$$\|\mathcal{A}X_{k+1} - b\|^2 = \frac{\|s_k\|^2}{\rho^2} \le \frac{2}{\rho}(h_k - h_{k+1} + \eta) \le \frac{2}{\rho}(h_k + \eta). \tag{4.29}$$

Substituting the bound on h_k proves $\|\mathcal{A}X_{k+1} - b\| \le r_k^{\mathrm{p}}$ without a multiplier boundedness assumption.

Raw primal objective. It remains to connect feasibility to the objective value of the same raw output. Write $r := \mathcal{A}X_{k+1} - b$. The support identity gives

$$m_k(y_{k+1}) = \mathscr{L}(X_{k+1}, y_{k+1}) = \langle C, X_{k+1}\rangle - \langle y_{k+1}, r\rangle.$$

The same model value equals $f^* + h_{k+1} - e_k$. Therefore

$$\langle C, X_{k+1}\rangle - f^* = h_{k+1} - e_k + \langle y_{k+1}, r\rangle. \tag{4.30}$$

The mixed accuracy estimate gives $h_{k+1} - e_k \ge 2h_{k+1} - h_k - 2\eta \ge -h_k - 2\eta$. If $M_H < \infty$, invariance gives $\|y_{k+1}\| \le M_H$, and hence

$$\langle C, X_{k+1}\rangle - f^* \ge -h_k - 2\eta - M_H\|r\|.$$

For the opposite direction, take $y^{\mathrm{ref}} = \mathrm{proj}_{\mathcal{Y}^*}(y_0)$. By Lemma 4.4, this multiplier belongs to $\mathcal{E}_0 \cap \mathcal{L}_H$ and has norm at most M_H. Since $X_{k+1} \in \mathcal{D}$,

$$\langle C, X_{k+1}\rangle - \langle y^{\mathrm{ref}}, r\rangle = \mathscr{L}(X_{k+1}, y^{\mathrm{ref}}) \le F(y^{\mathrm{ref}}) = f^*.$$

Thus $\langle C, X_{k+1}\rangle - f^* \le M_H\|r\|$. Combining both sides yields

$$|\langle C, X_{k+1}\rangle - f^*| \le h_k + 2\eta + M_H\|\mathcal{A}X_{k+1} - b\|. \tag{4.31}$$

Substituting (4.27) and the feasibility bound proves (4.28). □

The theorem distinguishes the quantities affected linearly and through a square root by the model tolerance. The objective gap and squared dual distance have $O(\eta)$ floors; the dual distance and raw primal errors have $O(\sqrt{\eta})$ floors. At $\eta = 0$, the floors vanish for any sequence of completed corrections. The proof also shows the precise role of boundedness: only the raw objective bound needs a uniform bound on the accepted multipliers and an optimal multiplier. Neither the dual estimates nor primal feasibility requires M_H.

4.4 Linear convergence under quadratic model accuracy

A fixed absolute tolerance leaves a nonzero error floor. For the local DD analysis, we will use a stronger accuracy property in which the model error decreases with the step itself:

$$e_k \le L\|s_k\|^2, \qquad L \ge 0, \qquad 2\rho L \le 1. \tag{4.32}$$

The constant L controls model error, not smoothness of F, and must be uniform over the steps being considered. The important feature is that the error can be absorbed by the negative squared-step term in (4.9). We first derive the resulting distance decrease without using quadratic growth; this estimate will later be used to establish local invariance.

Lemma 4.6 (Quadratic model accuracy and Fejér decrease)**.** *An exact correction satisfying* (4.32) *passes the relative test $e_k \le \Delta_k/2$ and, for every $y \in \mathcal{Y}^*$, satisfies*

$$\|y_{k+1} - y\|^2 \le \|y_k - y\|^2 - (1 - 2\rho L)\|s_k\|^2 - 2\rho h_{k+1} \le \|y_k - y\|^2. \tag{4.33}$$

Proof. By (4.7), $e_k \le L\|s_k\|^2 \le \|s_k\|^2/(2\rho) \le \Delta_k/2$. For $y \in \mathcal{Y}^*$, rearranging the three-point comparison gives

$$\|y_{k+1} - y\|^2 \le \|y_k - y\|^2 - \|s_k\|^2 - 2\rho h_{k+1} + 2\rho e_k.$$

Substitute $e_k \le L\|s_k\|^2$ to obtain the first inequality in (4.33). The second uses only $2\rho L \le 1$ and $h_{k+1} \ge 0$. □

The preceding lemma shows that quadratic model accuracy is sufficient for Fejér decrease and automatically implies the relative acceptance test. Adding quadratic growth converts this decrease into contraction of the distance to the optimal set. To conclude convergence to a *single* dual optimum, we will also establish summability of the step lengths.

Theorem 4.7 (Linear convergence under quadratic model accuracy)**.** *Suppose every exact correction satisfies* (4.32)*, and suppose there is a constant $\mu > 0$ such that $h_k \ge \mu d_k^2/2$ at every iterate. Set $q := (1 + \rho\mu)^{-1/2} < 1$. Then, for every $k \ge 0$,*

$$d_{k+1} \le q d_k, \qquad d_k \le d_0 q^k, \qquad h_{k+1} \le \frac{d_0^2}{2\rho} q^{2k}. \tag{4.34}$$

There exists $y_\infty \in \mathcal{Y}^*$ such that

$$\|s_k\| \le 2d_0 q^k, \qquad \|y_k - y_\infty\| \le \frac{2d_0}{1-q} q^k. \tag{4.35}$$

The raw primal equality residual satisfies

$$\|\mathcal{A}X_{k+1} - b\| \le \frac{2d_0}{\rho} q^k. \tag{4.36}$$

Thus the distance to $\mathcal{Y}^*$ contracts Q-linearly, while the dual iterates and primal equality residuals satisfy the stated R-linear bounds.

Proof. We first prove the distance and objective bounds, then use the step lengths to identify a limit of the dual sequence. Choose $y_k^{\mathrm{ref}} \in \mathcal{Y}^*$ nearest to y_k. Dropping the nonpositive squared-step term from Lemma 4.6 gives

$$d_{k+1}^2 + 2\rho h_{k+1} \le \|y_{k+1} - y_k^{\mathrm{ref}}\|^2 + 2\rho h_{k+1} \le d_k^2.$$

Quadratic growth gives $2\rho h_{k+1} \ge \rho\mu d_{k+1}^2$. Combining the two terms on the left proves the first inequality below; dropping the distance term instead proves the second:

$$(1 + \rho\mu) d_{k+1}^2 \le d_k^2, \qquad h_{k+1} \le \frac{d_k^2}{2\rho}. \tag{4.37}$$

Iteration proves (4.34).

The same Fejér estimate gives $\|y_{k+1} - y_k^{\mathrm{ref}}\| \le d_k$. Consequently,

$$\|s_k\| \le \|y_{k+1} - y_k^{\mathrm{ref}}\| + \|y_k - y_k^{\mathrm{ref}}\| \le 2d_k \le 2d_0 q^k.$$

These step lengths are summable, so (y_k) is Cauchy and converges to some y_∞. Since $d_k \to 0$ and $\mathcal{Y}^*$ is closed, $y_\infty \in \mathcal{Y}^*$. Summing the step bounds from k onward gives

$$\|y_k - y_\infty\| \le \sum_{j=k}^{\infty} \|s_j\| \le \frac{2d_0}{1-q} q^k.$$

This proves (4.35). Finally, $\mathcal{A}X_{k+1} - b = s_k/\rho$ yields (4.36). □

Theorem 4.7 isolates the two properties needed for local acceleration: quadratic model accuracy and quadratic growth. When these properties are only local, a separate invariance argument is essential. Lemma 4.6 can provide that argument without growth, but only after its model-error condition has been verified at the actual trial. Section 6 performs this verification for the DD construction, using local exactness and spectral estimates with $L = L_{\mathrm{sp}}$ or $L = L_{\mathrm{mix}}$.

The primal conclusion here concerns equality residuals. A rate for the matrix distance to a primal optimum needs stability of the primal correction map; it cannot be inferred from a residual bound alone. That additional step will also be addressed in the single-block DD analysis.

5 Convergence analysis of the diagonal algorithm

The common analysis gives rates for completed center updates. To prove the diagonal theorems, we must still show that the required model accuracy can be attained and count the QP solves used to attain it. The key observation is that, at an exact correction, the model error is the Frank–Wolfe gap of the full-domain correction objective. A rejected trial supplies a support atom for the next model, allowing that gap to be controlled by a fixed-center descent argument.

To connect notation with Section 4, let i_k be the accepted inner index in Algorithm 2.1, and set $\mathcal{K}_k := \mathcal{K}_{k,i_k}$, $m_k := m_{k,i_k}$, $e_k := e_{k,i_k}$, and $\Delta_k := \Delta_{k,i_k}$. The accepted output $X_{k+1} = X_{k,i_k+1}$ therefore satisfies (4.3). We first prove uniform and gap-dependent refinement bounds, and then combine them with the accepted-iterate estimates. All counts include rejected corrections.

5.1 Bounds on the number of inner iterations

Fixing the center leaves a single convex quadratic objective, even though the spectral models change. The next lemma shows that retaining the preceding correction and one new support atom suffices to reach any positive absolute model tolerance in a uniformly bounded number of QP solves.

Lemma 5.1 (Finite fixed-center refinement)**.** *At any center y_k, the inner loop with $\eta > 0$ terminates after at most*

$$n_k \le 2 + \frac{5C}{\eta} \tag{5.1}$$

QP solves, where $C = \rho D_{\mathcal{A}}^2$ is defined in (2.29).

Proof. The proof has three steps. We identify the model error with a full-domain linearization gap, use the refreshed model to obtain a descent comparison, and convert this comparison into a bound on the number of refinements.

Identification of the model error. Each model is a nonempty compact convex subset of $\mathcal{D}$, so an exact correction exists. Its gradient satisfies

$$\nabla \mathcal{J}_{y_k}(X_{k,i+1}) = \rho \mathcal{A}^*(\mathcal{A}X_{k,i+1} - b) - S(y_k) = -S(y_{k,i+1}), \tag{5.2}$$

and exact correction optimality makes $X_{k,i+1}$ a support maximizer on $\mathcal{K}_{k,i}$ at $y_{k,i+1}$. Consequently,

$$e_{k,i} = \max_{Z \in \mathcal{D}} \langle S(y_{k,i+1}), Z\rangle - \langle S(y_{k,i+1}), X_{k,i+1}\rangle = \max_{Z \in \mathcal{D}} \langle \nabla \mathcal{J}_{y_k}(X_{k,i+1}), X_{k,i+1} - Z\rangle. \tag{5.3}$$

Thus $e_{k,i}$ is the Frank–Wolfe gap for minimizing $\mathcal{J}_{y_k}$ over the full domain $\mathcal{D}$. This is stronger information than optimality within the current model and is what makes a rejected correction useful for refinement.

A descent comparison after refinement. Choose a full-domain support maximizer $U_{k,i}$ at the rejected trial $y_{k,i+1}$. If the largest eigenvalue is positive, take the trace-R atom corresponding to the first column of the refreshed basis; otherwise, take zero. The next model contains both this atom and $X_{k,i+1}$, and hence the segment joining them. For any $0 \le \alpha \le 1$, expansion of the quadratic objective gives

$$\mathcal{J}_{y_k}\big((1-\alpha)X_{k,i+1} + \alpha U_{k,i}\big) = \mathcal{J}_{y_k}(X_{k,i+1}) - \alpha e_{k,i} + \frac{\rho\alpha^2}{2}\|\mathcal{A}(U_{k,i} - X_{k,i+1})\|^2.$$

The squared residual term is at most $C\alpha^2/2$. Optimality of the next correction therefore implies

$$\mathcal{J}_{y_k}(X_{k,i+2}) \le \mathcal{J}_{y_k}(X_{k,i+1}) - \alpha e_{k,i} + \frac{C}{2}\alpha^2. \tag{5.4}$$

Notice that this argument uses only the two retained comparison points. No inclusion between successive spectral models is required.

From descent to the solve count. For the analysis only, continue the same fixed-center refinement even if a trial has already passed the test. Write $e_i := e_{k,i}$ and $H_i := \mathcal{J}_{y_k}(X_{k,i+1}) - \min_{X \in \mathcal{D}} \mathcal{J}_{y_k}(X)$. Convexity gives $0 \le H_i \le e_i$. Taking $\alpha = 1$ in the first comparison yields $H_1 \le H_0 - e_0 + C/2 \le C/2$. For $i \ge 1$, the choice $\alpha = 2/(i+2)$ in (5.4) gives, by induction,

$$H_i \le \frac{2C}{i+2} \qquad (i \ge 1). \tag{5.5}$$

Indeed, this holds at $i = 1$, and the induction step is

$$H_{i+1} \le \left(1 - \frac{2}{i+2}\right)\frac{2C}{i+2} + \frac{2C}{(i+2)^2} = \frac{2C(i+1)}{(i+2)^2} \le \frac{2C}{i+3}.$$

An objective-gap bound alone does not yet certify acceptance, since the test uses e_i. To control this larger quantity, fix $m \ge 1$, use $\alpha = 1/m$ for $i = m, \dots, 2m-1$, and sum the descent comparisons. We obtain

$$\min_{m \le i < 2m} e_i \le \frac{1}{m}\sum_{i=m}^{2m-1} e_i \le H_m - H_{2m} + \frac{C}{2m} \le \frac{2C}{m+2} + \frac{C}{2m} \le \frac{5C}{2m}. \tag{5.6}$$

If $C > 0$, choose $m = \max\{1, \lceil 5C/(2\eta)\rceil\}$. There is an index $i \in \{m, \dots, 2m-1\}$ with $e_i \le \eta$. The corresponding correction is among the first $2m$ QP solves. Hence the actual loop accepts no later, and may stop earlier through the relative test. Since $2m \le 2 + 5C/\eta$, the claimed bound follows. The continuation used to prove it is not executed by the algorithm. Finally, if $C = 0$, the first comparison gives $H_1 = 0$, and full-domain optimality implies $e_1 = 0$. Thus two QP solves suffice in this case as well. □

The refinement bound uses only one new support atom. Additional directions can improve the correction, but are not needed for finite termination, and no assumption on the optimal rank enters the proof.

The uniform bound depends on η. Under quadratic growth, a second bound relates refinement work to the current objective gap. This dependence is needed to obtain the improved total-work estimate without paying a uniform worst-case budget at every accepted center.

Lemma 5.2 (Gap-dependent refinement cost)**.** *Suppose $h_k := F(y_k) - f^* > 0$ and $h_k \geq \mu d_k^2/2$ for some $\mu > 0$. The relative test at this center is reached within*

$$2 + \frac{10C}{c_{\mathrm{QG}} h_k} \tag{5.7}$$

QP solves, unless the absolute test accepts an earlier trial.

Proof. By Lemma 4.4, every exact trial at this fixed center satisfies $\Delta_{k,i} \geq c_{\mathrm{QG}} h_k$, independently of whether it is accepted. Therefore, $e_{k,i} \leq c_{\mathrm{QG}} h_k/2$ suffices for the relative test. Apply the preceding fixed-center gap argument with the threshold $c_{\mathrm{QG}} h_k/2$. It gives $2 + 5C/(c_{\mathrm{QG}} h_k/2)$ solves, unless the algorithm has already accepted. The threshold is used only for the work bound; the algorithm neither changes its fixed tolerance nor needs to know h_k. □

5.2 Single-block guarantees and total QP work

We now have both components needed for the diagonal guarantees: Section 4 controls the accepted outputs, and the preceding lemmas control the work between them. The general convex result follows by direct composition. The quadratic-growth result requires a more careful sum of the gap-dependent costs.

Proof of Theorem 2.2. Every refinement loop terminates by Lemma 5.1. The accepted correction is then an instance of (4.3), and its acceptance test is exactly (4.10). Thus all assumptions of Theorem 4.3 hold. Taking $\Gamma_0 = \Gamma$ gives (2.30).

For a target $\epsilon > 0$, choose $\eta = \epsilon/4$ and $K = \max\{1, \lceil 2\Gamma/\epsilon \rceil\}$. Then $\Gamma/K + 2\eta \leq \epsilon$, so all three averaged errors are at most ϵ. Summing the refinement bound over these K accepted updates gives

$$N_{\mathrm{QP}} = \sum_{k=0}^{K-1} n_k \leq K \left(2 + \frac{5C}{\eta}\right) = K \left(2 + \frac{20C}{\epsilon}\right) = O(\epsilon^{-2}).$$

This proves (2.31). □

Proof of Theorem 2.3. The accepted-iterate estimates follow from the common theorem. The main additional task is to prove the total QP count. We first bound the work up to the first small-gap center, then take one further correction to obtain the required raw primal output.

Accepted-iterate bounds. Apply Theorem 4.5 to the accepted support-model sequence. It gives sublevel invariance, the dual estimates, and, under the multiplier bound, the raw primal estimates. Taking $q = q_{\mathrm{mix}}$ and replacing k by $K - 1$ gives (2.36) and, under the multiplier bound, (2.37), with the constants in (2.34).

First target hit. Set $c := c_{\mathrm{QG}}$ and $\theta := 1 - c/2 \in (0, 1)$. Fix $0 < \tau \leq H$ and run the algorithm with the fixed tolerance

$$\eta = \frac{c\tau}{4}. \tag{5.8}$$

Since $\eta \leq cH/2$, Lemma 4.4 keeps every accepted center in $\mathcal{E}_0 \cap \mathcal{L}_H$. Let $J := \inf\{k \geq 0 : h_k \leq \tau\}$, allowing $J = +\infty$ temporarily. For every $k < J$, Lemma 4.4 gives $\Delta_k \geq c h_k > c\tau = 4\eta$. An accepted trial with $e_k \leq \eta$ therefore also satisfies $e_k < \Delta_k/4 \leq \Delta_k/2$. Hence every pre-hit update is relative, and the accepted-step decrease gives

$$h_{k+1} = h_k - \Delta_k + e_k \leq h_k - \frac{\Delta_k}{2} \leq \theta h_k \qquad (0 \leq k < J). \tag{5.9}$$

Every refinement loop is finite by Lemma 5.1. If J were infinite, (5.9) would imply $h_k \le \theta^k h_0 \to 0$, contradicting $h_k > \tau$ for every k. Thus $J < \infty$.

The cost at a pre-hit center is proportional to $1/h_k$, so we must sum these reciprocals rather than just count accepted steps. If $J \ge 1$, the contraction gives $h_{J-1} \le \theta^{J-1-k} h_k$ for $0 \le k \le J-1$. Since $h_{J-1} > \tau$, all denominators are positive and

$$\sum_{k=0}^{J-1} \frac{1}{h_k} \le \frac{1}{h_{J-1}} \sum_{k=0}^{J-1} \theta^{J-1-k} = \frac{1-\theta^J}{(1-\theta)h_{J-1}} < \frac{1}{(1-\theta)\tau}. \tag{5.10}$$

The final pre-hit gap is used here because h_J may be zero. Also $h_k \le h_0$ for $k < J$, so $J \le h_0 \sum_{k=0}^{J-1} h_k^{-1}$. Summing Lemma 5.2 consequently yields

$$N_J := \sum_{k=0}^{J-1} n_k \le 2J + \frac{10C}{c} \sum_{k=0}^{J-1} \frac{1}{h_k} \le \frac{2h_0 + 10C/c}{(1-\theta)\tau}. \tag{5.11}$$

If $h_0 \le \tau$, then $J = N_J = 0$, so the bound remains valid. Taking $\tau = \epsilon$ proves the $O(\epsilon^{-1})$ work bound for attaining a dual gap at most ϵ. The important point is the reverse geometric sum in (5.10): assigning the same worst-case refinement budget to every accepted step would introduce an unnecessary logarithmic factor.

A terminal raw primal output. The small gap at y_J does not yet give the desired bound for the primal matrix that produced this center: the raw residual is controlled by the gap at the *starting* center of its correction. We therefore take one additional accepted correction from y_J. Its cost satisfies

$$n_J \le 2 + \frac{5C}{\eta} = 2 + \frac{20C}{c\tau}. \tag{5.12}$$

This uniform bound remains valid when $h_J = 0$. By (4.12), $h_{J+1} \le h_J + \eta \le (1 + c/4)\tau$. Growth at y_{J+1} and the raw residual estimate (4.29) then give

$$\begin{aligned} h_{J+1} &\le (1 + c/4)\tau, \\ \operatorname{dist}(y_{J+1}, \mathcal{Y}^*) &\le \sqrt{\frac{2(1 + c/4)\tau}{\mu}}, \\ \|\mathcal{A}X_{J+1} - b\| &\le \sqrt{\frac{2(h_J + \eta)}{\rho}} \le \sqrt{\frac{2(1 + c/4)\tau}{\rho}}. \end{aligned} \tag{5.13}$$

These bounds now concern the same terminal pair (X_{J+1}, y_{J+1}), with primal feasibility controlled by the small-gap starting center y_J.

For the objective error of this terminal output, we can obtain a multiplier bound directly, without assuming that the entire sublevel is bounded. Suppose also that $\tau \le 1$, fix an initial nearest optimum y^{ref}, and let $W_k := \|y_k - y^{\mathrm{ref}}\|^2 + 2\rho h_k$, so $W_0 = E_0$. Every pre-hit step is relative. Hence the fixed-reference version of Lemma 4.2 gives

$$W_{k+1} \le W_k - \|s_k\|^2 - 2\rho h_{k+1} \le W_k \qquad (k < J),$$

and hence $W_J \le E_0$. This is also true for $J = 0$. The additional mixed step satisfies

$$W_{J+1} \le W_J - \|s_J\|^2 - 2\rho h_{J+1} + 4\rho\eta \le E_0 + 4\rho\eta \le E_0 + \rho c.$$

Thus $M_0 := \|y^{\mathrm{ref}}\| + \sqrt{E_0 + \rho c}$ bounds both $\|y_{J+1}\|$ and $\|y^{\mathrm{ref}}\|$, independently of τ.

Let $r_J := \mathcal{A}X_{J+1} - b$. The exact support identity (4.30) and the terminal accuracy bound imply

$$\langle C, X_{J+1}\rangle - f^* = h_{J+1} - e_J + \langle y_{J+1}, r_J\rangle \ge 2h_{J+1} - h_J - 2\eta - M_0\|r_J\| \ge -h_J - 2\eta - M_0\|r_J\|.$$

On the other hand, $\mathscr{L}(X_{J+1}, y^{\mathrm{ref}}) \le F(y^{\mathrm{ref}}) = f^*$ gives $\langle C, X_{J+1}\rangle - f^* \le \langle y^{\mathrm{ref}}, r_J\rangle \le M_0\|r_J\|$. Combining the two sides and substituting (5.13) yields

$$|\langle C, X_{J+1}\rangle - f^*| \le (1 + c/2)\tau + M_0\sqrt{\frac{2(1 + c/4)\tau}{\rho}}. \tag{5.14}$$

Thus the terminal dual distance, primal residual, and absolute primal objective error are all $O(\sqrt{\tau})$ for $0 < \tau \le 1$, with constants independent of the target τ.

Total work and target selection. Combining (5.11) and (5.12) bounds all work through this output by

$$\sum_{k=0}^{J} n_k \le \frac{2h_0 + 10C/c}{(1-\theta)\tau} + 2 + \frac{20C}{c\tau} = O(\tau^{-1}). \tag{5.15}$$

For sufficiently small $\epsilon > 0$, choose $\tau = \epsilon/(1 + c/4)$ and $\eta = c\tau/4$. The same terminal pair then has dual gap at most ϵ and dual-distance and raw primal errors of order $\sqrt{\epsilon}$, using $O(\epsilon^{-1})$ QP solves. Alternatively, taking τ proportional to ϵ^2, with a sufficiently small fixed proportionality constant, makes all three latter errors at most ϵ in $O(\epsilon^{-2})$ QP solves. This proves the finite-accuracy complexity assertions. □

The proof uses the first-hit index to certify that a suitable output is attained within the stated work budget. It does not require the algorithm to evaluate f^* or recognize that index. The actual acceptance decision uses only $\Delta_{k,i}$ and $e_{k,i}$. There is also a useful distinction between the terminal and all-iterate bounds: quadratic growth on the relevant sublevel suffices for the finite-target argument above, whereas a uniform multiplier bound is needed for the all-iterate raw objective estimate in Theorem 4.5.

5.3 Proof of the multi-cone diagonal guarantee

To extend the preceding argument, we need to verify that a refreshed product model contains the same two comparison points used in the single-block descent estimate. Once this is established, the work bounds depend on the product-domain curvature and not on how many blocks are represented exactly.

Proof of Theorem 3.1(i). Support maximization on (3.2) separates by block. At the common trial multiplier, select a support maximizer on each exact block and a leading rank-one atom, or zero, on each large PSD block. Together these form a full-domain support maximizer $U_{k,i}$. The refreshed product model contains $U_{k,i}$ because each large block retains its new atom, and it contains $X_{k,i+1}$ because each large block retains the preceding output. Exact blocks retain their whole sets. Convexity therefore places the segment between these two points in the next model.

Define $D_{\mathcal{A}} := \sup_{X,Z\in\mathcal{D}} \|\mathcal{A}(X - Z)\|$ and $C := \rho D_{\mathcal{A}}^2$. For an exact joint correction, its model error is the complete Frank–Wolfe gap, and the quadratic remainder along the retained segment is at most $C\alpha^2/2$. The proofs of Lemmas 5.1 and 5.2 therefore give the same uniform and gap-dependent refinement bounds, now counting joint model solves.

The support-model results of Section 4 apply to this compact convex product domain. Consequently, Theorem 4.3 gives (3.19); choosing $\eta = \epsilon/4$ and summing the uniform refinement costs gives $O(\epsilon^{-2})$ work. If the complete penalty satisfies quadratic growth on the relevant sublevel containing the initial center, the same tuned tolerance and first-hit argument leading to (5.11) give $O(\epsilon^{-1})$ work for a current-center penalty gap at most ϵ. The terminal estimates (5.13)– (5.15) also hold for the product primal output X_{J+1}.

This proves part (i). The argument uses support separability, not separability of the quadratic correction: $\mathcal{A}X - b$ still couples the optimization variables. The exact blocks appear in the product domain and its curvature constant, and the improved bound requires quadratic growth of the complete penalty. All work bounds count exact joint convex quadratic conic corrections, not the numerical iterations used to solve them. □

6 Convergence analysis of the DD algorithm

Throughout this section, the hypotheses of Theorem 2.5 hold. The arguments through local DD exactness do not use quadratic growth; Assumption 2.2(ii) is used only for the final rate. The SDP data, the selected optimal pair $(X_\star, y_\star)$, the ranks s, r, and the parameters ρ, ε are fixed. All constants and neighborhoods may depend on these fixed objects and the stated quadratic-growth constants, but not on the iteration index or the particular admissible ordered eigenbasis. In particular, no uniformity as $\varepsilon \downarrow 0$ is asserted.

Proof roadmap. The proof has two parts.

(i) *Local equality of the QP and an auxiliary active-space problem.* Let $P_s(y)$ be the leading rank-s spectral projector of $S(y)$ and set

$$\begin{aligned} \mathcal{D}_s(y) &= \{X \in \mathcal{D} : X = P_s(y) X P_s(y)\}, \\ X(y) &= \operatorname*{argmin}_{X \in \mathcal{D}_s(y)} \left\{ \mathcal{J}_y(X) = \frac{\rho}{2}\|\mathcal{A}(X) - b\|^2 - \langle S(y), X\rangle \right\}. \end{aligned} \tag{6.1}$$

We prove that, in a joint neighborhood of $(y_\star, X_\star)$, the actual DD-QP has the same unique matrix solution $X(y)$. Consequently, with $V_{k,s}$ the first s columns of the ordered working basis V_k, its dual update is exactly

$$\begin{aligned} m_{V_{k,s}}(z) &= b^\top z + \max_{X \in \mathcal{D}_s(y_k)} \langle S(z), X\rangle \\ &= b^\top z + R \max\{0, \lambda_1(V_{k,s}^\top S(z) V_{k,s})\}, \\ y_{k+1} &= \operatorname*{argmin}_{z \in \mathbb{R}^m} \left\{ m_{V_{k,s}}(z) + \frac{\|z - y_k\|^2}{2\rho} \right\} = y_k + \rho\big(\mathcal{A}(X(y_k)) - b\big). \end{aligned} \tag{6.2}$$

The auxiliary problem is used only in the analysis; the algorithm solves only its coefficient QP. When $r > s$, the comparison set is $\mathcal{D}_s(y_k)$, not the full r-dimensional PSD trace ball.

(ii) *Second-order model accuracy and proximal contraction.* The active spectral model satisfies, for nearby y, z and $V_s \in \mathrm{TopEig}_s(S(y))$,

$$0 \le F(z) - m_{V_s}(z) \le L_{\mathrm{sp}}\|z - y\|^2.$$

The proximal optimality inequality and this second-order error estimate first keep the primal–dual state in the joint neighborhood. Quadratic growth then gives a linear contraction of the distance to $\mathcal{Y}^*$, and the Lipschitz continuity of $X(y)$ gives the primal and residual rates. Thus the argument follows the standard second-order model, proximal descent, and quadratic-growth mechanism.

6.1 Local spectral structure at the optimum

Throughout this section, the hypotheses of the DD convergence theorem are in force. In particular, $X_\star$ is the unique primal optimum, $\operatorname{tr} X_\star < R$, $s = \operatorname{rank} X_\star \in \{1, \dots, n-1\}$, and $Z_\star = \mathcal{A}^* y_\star - C = -S(y_\star)$ satisfies $X_\star Z_\star = 0$ and $X_\star + Z_\star \succ 0$.

Lemma 6.1 (Strict complementarity gives the hidden active gap)**.** *There exist $Q_\star \in O(n)$ and positive diagonal matrices $\Sigma_\star \in \mathbb{S}^s_{++}$ and $\Lambda_\star \in \mathbb{S}^{n-s}_{++}$ such that*

$$X_\star = Q_\star \begin{pmatrix} \Sigma_\star & 0 \\ 0 & 0 \end{pmatrix} Q_\star^\top, \qquad Z_\star = Q_\star \begin{pmatrix} 0 & 0 \\ 0 & \Lambda_\star \end{pmatrix} Q_\star^\top. \tag{6.3}$$

Define the positive margins

$$\sigma_\star = \lambda_{\min}(\Sigma_\star), \qquad \gamma_\star = \lambda_{\min}(\Lambda_\star). \tag{6.4}$$

If $P_\star$ is the orthogonal projector onto range $X_\star$*, then*

$$\ker Z_\star = \operatorname{range} X_\star, \qquad Z_\star \succeq \gamma_\star(I - P_\star), \qquad \lambda_s(S(y_\star)) = 0, \quad \lambda_{s+1}(S(y_\star)) = -\gamma_\star. \tag{6.5}$$

Proof. The symmetric matrices $X_\star$ and $Z_\star$ commute, since $X_\star Z_\star = Z_\star X_\star = 0$. They therefore admit a common orthonormal eigenbasis. Choose its matrix $Q_\star \in O(n)$ with columns ordered by nonincreasing eigenvalues of $X_\star$. Its first s eigenvalues are positive and the remaining ones are zero. Complementarity forces the first s eigenvalues of $Z_\star$ in this basis to be zero. Strict complementarity forces all remaining eigenvalues of $Z_\star$ to be positive. This gives (6.3). Reading the kernels and eigenvalues from these diagonal forms, and using $S(y_\star) = -Z_\star$, gives (6.5). □

Remark 6.2. Let $U_\star$ consist of the first s columns of $Q_\star$. Then

$$X_\star = U_\star \Sigma_\star U_\star^\top, \qquad U_\star \in \mathrm{TopEig}_s(S(y_\star)), \qquad P_\star = U_\star U_\star^\top.$$

The positive margin $\sigma_\star$ controls the primal active block; $\gamma_\star$ controls the spectral separation from the inactive block. No gap is required at the working rank r.

For a symmetric matrix G with $\lambda_s(G) > \lambda_{s+1}(G)$, let $P_s(G)$ denote its unique leading rank-s spectral projector. We abbreviate $P_s(y) = P_s(S(y))$ whenever this gap is positive.

Lemma 6.3 (Uniform active gap, spectral width, and projector variation)**.** *Choose $R_Y > 0$ so that*

$$\|\mathcal{A}\| R_Y \le \frac{\gamma_\star}{16}, \tag{6.6}$$

and let $\overline{B}(y_\star, R_Y)$ be the closed Euclidean ball. Set

$$\underline{\delta} = \frac{7}{8}\gamma_\star, \qquad \overline{\Gamma} = \max_{y \in \overline{B}(y_\star, R_Y)} \big(\lambda_1(S(y)) - \lambda_n(S(y))\big), \qquad L_P = \frac{\sqrt{2}\|\mathcal{A}\|}{\underline{\delta}}. \tag{6.7}$$

Then $\overline{\Gamma} < \infty$ and, for all y, z in this ball,

$$\lambda_s(S(y)) - \lambda_{s+1}(S(y)) \ge \underline{\delta}, \tag{6.8}$$

$$\lambda_1(S(y)) - \lambda_n(S(y)) \le \overline{\Gamma}, \tag{6.9}$$

$$\|P_s(y) - P_s(z)\|_F \le L_P \|y - z\|_2. \tag{6.10}$$

Proof. The adjoint $\mathcal{A}^*$ maps the Euclidean norm on $\mathbb{R}^m$ to the Frobenius norm on $\mathbb{S}^n$, with induced norm $\|\mathcal{A}^*\| = \|\mathcal{A}\|$. Hence

$$\|S(y) - S(z)\|_2 \le \|S(y) - S(z)\|_F \le \|\mathcal{A}\| \|y - z\|_2.$$

By Weyl's inequality and Lemma 6.1,

$$\lambda_s(S(y)) \ge -\|\mathcal{A}\| R_Y, \qquad \lambda_{s+1}(S(y)) \le -\gamma_\star + \|\mathcal{A}\| R_Y.$$

Their difference is at least $\gamma_\star - 2\|\mathcal{A}\| R_Y \ge \underline{\delta}$. Continuity of the eigenvalues and compactness of the ball show that the defining maximum for $\overline{\Gamma}$ is finite.

The Davis–Kahan $\sin\Theta$ theorem [51] applied to the leading s-dimensional spectral subspaces gives

$$\|P_s(y) - P_s(z)\|_F \le \frac{\sqrt{2}}{\underline{\delta}} \|S(y) - S(z)\|_F \le L_P \|y - z\|.$$

Indeed, every leading eigenvalue at either center and every inactive eigenvalue at the other are separated by at least $\gamma_\star - 2\|\mathcal{A}\| R_Y \ge \underline{\delta}$. This is precisely the separation required for the stated projector form of the theorem, and proves (6.10). □

Lemma 6.4 (Equality residuals control active perturbations)**.** *There is a constant $\alpha_\star > 0$ such that*

$$\|\mathcal{A}(U_\star Z U_\star^\top)\|_2 \ge \alpha_\star \|Z\|_F \quad \text{for every } Z \in \mathbb{S}^s, \qquad \alpha_\star = \min_{\substack{Z \in \mathbb{S}^s \\ \|Z\|_F = 1}} \|\mathcal{A}(U_\star Z U_\star^\top)\|_2. \tag{6.11}$$

Proof. Suppose $Z \neq 0$ satisfies $\mathcal{A}(U_\star Z U_\star^\top) = 0$. Since $\Sigma_\star \succ 0$ and $\operatorname{tr} X_\star < R$, for all sufficiently small $t > 0$ the two distinct matrices

$$X_\pm = X_\star \pm t U_\star Z U_\star^\top = U_\star(\Sigma_\star \pm tZ)U_\star^\top$$

are positive semidefinite and have trace strictly less than R. They also satisfy $\mathcal{A}(X_\pm) = b$. The equality $C = \mathcal{A}^* y_\star - Z_\star$ and $Z_\star U_\star = 0$ imply

$$\langle C, U_\star Z U_\star^\top \rangle = y_\star^\top \mathcal{A}(U_\star Z U_\star^\top) - \langle U_\star^\top Z_\star U_\star, Z \rangle = 0.$$

Thus both X_+ and X_- are primal optima, contrary to uniqueness. The linear map $Z \mapsto \mathcal{A}(U_\star Z U_\star^\top)$ is therefore injective. Its norm is a continuous positive function on the compact unit sphere of $\mathbb{S}^s$, so its minimum $\alpha_\star$ is positive. Homogeneity gives the inequality for arbitrary Z. □

Lemma 6.5 (Local stability within the dual solution set)**.** *Let $t_\star = R - \operatorname{tr} X_\star > 0$ and $\mathcal{Y}^* = \operatorname{argmin} F$. For every $y \in \mathbb{R}^m$,*

$$\begin{aligned} F(y) - f^* &= R[\lambda_1(S(y))]_+ - \langle S(y), X_\star \rangle \\ &= t_\star[\lambda_1(S(y))]_+ + \langle [\lambda_1(S(y))]_+ I - S(y), X_\star \rangle \geq t_\star[\lambda_1(S(y))]_+. \end{aligned} \tag{6.12}$$

Every $z \in \mathcal{Y}^*$ satisfies $S(z) \preceq 0$ and $S(z)P_\star = 0$. If also $\|\mathcal{A}\|\|z - y_\star\|_2 \leq \gamma_\star/2$, then

$$-S(z) \geq \frac{\gamma_\star}{2}(I - P_\star), \qquad \ker S(z) = \operatorname{range} X_\star. \tag{6.13}$$

Proof. Primal feasibility gives $\langle S(y), X_\star \rangle = \langle C, X_\star \rangle - b^\top y$. Since $f^* = \langle C, X_\star \rangle$, substitution in the definition of F proves the first equality of (6.12). Adding and subtracting $[\lambda_1(S(y))]_+ \operatorname{tr} X_\star$ gives the second. The matrix $[\lambda_1(S(y))]_+ I - S(y)$ is positive semidefinite, so its inner product with $X_\star \succeq 0$ is nonnegative.

For $z \in \mathcal{Y}^*$, the left side of (6.12) vanishes. Since $t_\star > 0$, we obtain $[\lambda_1(S(z))]_+ = 0$, hence $S(z) \preceq 0$, and $\langle -S(z), X_\star \rangle = 0$. For positive semidefinite matrices B, X, the identity $\operatorname{tr}(BX) = \|B^{1/2}X^{1/2}\|_F^2$ shows that $\operatorname{tr}(BX) = 0$ implies $BX = 0$. Applying this with $B = -S(z)$ yields $S(z)X_\star = 0$, and thus $S(z)P_\star = 0$ because $X_\star$ is invertible on its range. For any $v \perp \operatorname{range} X_\star$, Lemma 6.1 now gives

$$\begin{aligned} v^\top[-S(z)]v &\geq (\gamma_\star - \|S(z) - S(y_\star)\|_2)\,\|v\|_2^2 \\ &\geq (\gamma_\star - \|\mathcal{A}\|\|z - y_\star\|_2)\,\|v\|_2^2 \geq \frac{\gamma_\star}{2}\|v\|_2^2. \end{aligned}$$

The active and cross blocks vanish because $S(z)P_\star = 0$. This proves both assertions in (6.13). □

6.2 The auxiliary minimizer on the active space

We next analyze the auxiliary problem on the full s-dimensional active space. A continuously varying orthonormal frame lets us compare its solutions in a fixed feasible set. Only Lipschitz continuity is needed; no differentiability of individual eigenvectors is assumed.

Lemma 6.6 (Stability of inverses and inverse square roots)**.** *For invertible matrices A, B,*

$$A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}.$$

If $M, N \succeq cI$ with $c > 0$, then

$$\|M^{-1/2} - N^{-1/2}\|_F \leq \frac{1}{2c^{3/2}}\|M - N\|_F. \tag{6.14}$$

Proof. The first identity follows by multiplication. For the second, set $A = M^{1/2}$, $B = N^{1/2}$, and $E = A - B$. Although A and B need not commute, we have

$$AE + EB = A(A - B) + (A - B)B = A^2 - B^2 = M - N.$$

Choose orthogonal diagonalizations $A = Q\,\mathrm{diag}(a_i)Q^\top$ and $B = R\,\mathrm{diag}(b_j)R^\top$. Multiplication on the left by $Q^\top$ and on the right by R gives

$$(a_i + b_j)(Q^\top E R)_{ij} = (Q^\top (M - N)R)_{ij}.$$

Since $a_i, b_j \geq \sqrt{c}$, summing the squared entries yields $\|A - B\|_F \leq \|M - N\|_F/(2\sqrt{c})$. Apply the inverse identity and $\|A^{-1}\|_{\mathrm{op}}, \|B^{-1}\|_{\mathrm{op}} \leq c^{-1/2}$ to obtain (6.14). □

Let R_Y and L_P be as in Lemma 6.3, and define

$$R_0 := \min\left\{R_Y, \frac{1}{4(1 + L_P)}\right\}, \qquad L_U := 2\sqrt{2}L_P. \tag{6.15}$$

For $y \in \overline{B}(y_\star, R_0)$, set

$$C_y := U_\star^\top P_s(y) U_\star, \qquad U(y) := P_s(y) U_\star C_y^{-1/2}. \tag{6.16}$$

Proposition 6.7 (An explicit local orthonormal frame)**.** *For every $y \in \overline{B}(y_\star, R_0)$,*

$$\tfrac{3}{4} I_s \preceq C_y \preceq I_s, \qquad U(y) \in \mathrm{St}(n, s), \qquad \mathrm{range}\, U(y) = \mathrm{range}\, P_s(y), \qquad U(y_\star) = U_\star.$$

Moreover, for all $y, z \in \overline{B}(y_\star, R_0)$,

$$\|U(y) - U(z)\|_F \leq L_U \|y - z\|. \tag{6.17}$$

Proof. Since $C_{y_\star} = I_s$,

$$\|C_y - I_s\|_{\mathrm{op}} \leq \|P_s(y) - P_\star\|_{\mathrm{op}} \leq L_P \|y - y_\star\| \leq \tfrac{1}{4}.$$

This proves the lower bound; the upper bound follows from $0 \preceq P_s(y) \preceq I$. The projector identity $P_s(y)^2 = P_s(y)$ gives

$$U(y)^\top U(y) = C_y^{-1/2} U_\star^\top P_s(y) U_\star C_y^{-1/2} = I_s.$$

The s orthonormal columns belong to the s-dimensional range of $P_s(y)$, so they span it. The formula at $y_\star$ is immediate.

For the Lipschitz estimate, the weaker bound $C_y, C_z \succeq I_s/2$ and Lemma 6.6 give

$$\|C_y^{-1/2}\|_{\mathrm{op}} \leq \sqrt{2}, \qquad \|C_y^{-1/2} - C_z^{-1/2}\|_F \leq \sqrt{2}\|C_y - C_z\|_F.$$

Expanding the difference in (6.16) as

$$(P_s(y) - P_s(z))U_\star C_y^{-1/2} + P_s(z)U_\star(C_y^{-1/2} - C_z^{-1/2})$$

and using $\|C_y - C_z\|_F \leq \|P_s(y) - P_s(z)\|_F$ bounds its Frobenius norm by $2\sqrt{2}\|P_s(y) - P_s(z)\|_F$. Lemma 6.3 proves (6.17). □

The frame $U(y)$ need not diagonalize $S(y)$ column by column. Its role is to give fixed coordinates for the entire active space. The algorithm continues to use an ordered eigenframe from $\mathrm{TopEig}_r(S(y))$.

Lemma 6.8 (Uniform injectivity on nearby active spaces)**.** *Let $\alpha_\star > 0$ be the constant in Lemma 6.4 and define*

$$R_1 := \min\left\{R_0, \frac{\alpha_\star}{4\|\mathcal{A}\| L_U}\right\}. \tag{6.18}$$

Then, for every $y \in \overline{B}(y_\star, R_1)$ and $Z \in \mathbb{S}^s$,

$$\|\mathcal{A}\big(U(y) Z U(y)^\top\big)\| \geq \frac{\alpha_\star}{2}\|Z\|_F. \tag{6.19}$$

Proof. For any $y, z \in \overline{B}(y_\star, R_0)$,

$$\begin{aligned} U(y)ZU(y)^\top - U(z)ZU(z)^\top &= (U(y) - U(z))ZU(y)^\top \\ &\quad + U(z)Z(U(y) - U(z))^\top. \end{aligned}$$

Consequently,

$$\|U(y)ZU(y)^\top - U(z)ZU(z)^\top\|_F \le 2L_U\|y - z\|\|Z\|_F. \tag{6.20}$$

Apply the triangle inequality, Lemma 6.4, and (6.20) with $z = y_\star$ to obtain

$$\begin{aligned} \|\mathcal{A}\big(U(y)ZU(y)^\top\big)\| &\ge \|\mathcal{A}(U_\star Z U_\star^\top)\| - \|\mathcal{A}\|\|U(y)ZU(y)^\top - U_\star Z U_\star^\top\|_F \\ &\ge \big(\alpha_\star - 2\|\mathcal{A}\|L_U R_1\big)\|Z\|_F \ge \frac{\alpha_\star}{2}\|Z\|_F. \end{aligned}$$

□

Let $\Theta_s := \{W \in \mathbb{S}^s : W \succeq 0,\ \operatorname{tr} W \le R\}$. In the frame (6.16), the auxiliary feasible set and objective are

$$\mathcal{D}_s(y) = \{U(y)WU(y)^\top : W \in \Theta_s\}, \qquad q_y(W) := \mathcal{J}_y\big(U(y)WU(y)^\top\big), \tag{6.21}$$

where explicitly

$$q_y(W) = \frac{\rho}{2}\|\mathcal{A}\big(U(y)WU(y)^\top\big) - b\|^2 - \langle S(y), U(y)WU(y)^\top\rangle. \tag{6.22}$$

The set Θ_s is independent of y, which is essential for the comparison below.

Theorem 6.9 (Existence, uniqueness, and Lipschitz continuity of the auxiliary minimizer)**.** *For every $y \in \overline{B}(y_\star, R_1)$, the problem*

$$W(y) := \operatorname*{argmin}_{W \in \Theta_s} q_y(W) \tag{6.23}$$

has a unique solution. It satisfies $W(y_\star) = \Sigma_\star$ and

$$\|W(y) - W(z)\|_F \le L_W\|y - z\| \qquad \big(y, z \in \overline{B}(y_\star, R_1)\big), \tag{6.24}$$

for some finite constant $L_W > 0$.

Proof. (1) Strong convexity, existence, and uniqueness. For $W, Z \in \mathbb{S}^s$, differentiation along $W + tZ$ gives

$$\begin{aligned} Dq_y(W)[Z] &= \rho\langle\mathcal{A}\big(U(y)WU(y)^\top\big) - b, \mathcal{A}\big(U(y)ZU(y)^\top\big)\rangle \\ &\quad - \langle S(y), U(y)ZU(y)^\top\rangle. \end{aligned} \tag{6.25}$$

Expanding the squared residual exactly and applying (6.19) yields

$$\begin{aligned} q_y(W + Z) - q_y(W) - Dq_y(W)[Z] &= \frac{\rho}{2}\|\mathcal{A}\big(U(y)ZU(y)^\top\big)\|^2 \\ &\ge \frac{\rho\alpha_\star^2}{8}\|Z\|_F^2. \end{aligned} \tag{6.26}$$

Thus q_y is strongly convex on $\mathbb{S}^s$, with modulus $\rho\alpha_\star^2/4$. Since Θ_s is nonempty, compact, and convex, continuity gives existence and strong convexity gives uniqueness.

(2) Identification at $y_\star$. The relations $S(y_\star)U_\star = 0$ and $U(y_\star) = U_\star$ imply $q_{y_\star}(W) \ge 0$ for all $W \in \Theta_s$. On the other hand, $\Sigma_\star \in \Theta_s$ and $\mathcal{A}(U_\star\Sigma_\star U_\star^\top) = \mathcal{A}(X_\star) = b$, so $q_{y_\star}(\Sigma_\star) = 0$. Uniqueness proves $W(y_\star) = \Sigma_\star$.

(3) Lipschitz continuity by comparing two optimality conditions. Take $y, z \in \overline{B}(y_\star, R_1)$ and write $W_y = W(y)$, $W_z = W(z)$, and $\Delta W = W_y - W_z$. The first-order optimality conditions on the same convex set Θ_s give

$$Dq_y(W_y)[W_z - W_y] \ge 0, \qquad Dq_z(W_z)[W_y - W_z] \ge 0.$$

Subtracting the derivative at W_z for the parameter y therefore gives

$$\begin{aligned}\frac{\rho\alpha_\star^2}{4}\|\Delta W\|_F^2 &\le \rho\|\mathcal{A}\big(U(y)\Delta W U(y)^\top\big)\|^2\\ &= (Dq_y(W_y) - Dq_y(W_z))[\Delta W]\\ &\le (Dq_z(W_z) - Dq_y(W_z))[\Delta W].\end{aligned} \tag{6.27}$$

The following finite constants will be used in the derivative estimate:

$$\begin{aligned}K_S &:= \|S(y_\star)\|_{\mathrm{op}} + \|\mathcal{A}\|R_0,\\ L_{\mathrm{der}} &:= 2\rho\|\mathcal{A}\|L_U(2R\|\mathcal{A}\| + \|b\|) + \|\mathcal{A}\| + 2K_S L_U,\\ L_W &:= \frac{4L_{\mathrm{der}}}{\rho\alpha_\star^2}.\end{aligned} \tag{6.28}$$

It remains to bound the change of the derivative at the same matrix. For $W \in \Theta_s$, we have $\|W\|_F \le R$ and hence

$$\begin{aligned}\|\mathcal{A}\big(U(y)WU(y)^\top\big) - b\| &\le R\|\mathcal{A}\| + \|b\|,\\ \|\mathcal{A}\big(U(y)WU(y)^\top - U(z)WU(z)^\top\big)\| &\le 2R\|\mathcal{A}\|L_U\|y - z\|.\end{aligned}$$

For a direction $Z \in \mathbb{S}^s$,

$$\begin{aligned}\|\mathcal{A}\big(U(y)ZU(y)^\top\big)\| &\le \|\mathcal{A}\|\|Z\|_F,\\ \|\mathcal{A}\big(U(y)ZU(y)^\top - U(z)ZU(z)^\top\big)\| &\le 2\|\mathcal{A}\|L_U\|y - z\|\|Z\|_F.\end{aligned}$$

Add and subtract one intermediate inner product in the quadratic part of (6.25). The preceding four bounds show that its change is at most

$$2\rho\|\mathcal{A}\|L_U(2R\|\mathcal{A}\| + \|b\|)\|y - z\|\|Z\|_F.$$

For the linear part, write

$$\begin{aligned}&\langle S(y), U(y)ZU(y)^\top\rangle - \langle S(z), U(z)ZU(z)^\top\rangle\\ &= \langle S(y) - S(z), U(y)ZU(y)^\top\rangle\\ &\quad + \langle S(z), (U(y) - U(z))ZU(y)^\top\rangle + \langle S(z), U(z)Z(U(y) - U(z))^\top\rangle.\end{aligned}$$

The first term is bounded in absolute value by $\|\mathcal{A}\|\|y - z\|\|Z\|_F$. For the second term, cyclicity of the trace and Cauchy–Schwarz give

$$\big|\langle S(z), (U(y) - U(z))ZU(y)^\top\rangle\big| \le \|S(z)\|_{\mathrm{op}}\|U(y) - U(z)\|_F\|Z\|_F.$$

The third term has the same bound. Because $\|S(z)\|_{\mathrm{op}} \le K_S$, these estimates prove

$$\big|Dq_y(W)[Z] - Dq_z(W)[Z]\big| \le L_{\mathrm{der}}\|y - z\|\|Z\|_F \qquad (W \in \Theta_s,\ Z \in \mathbb{S}^s). \tag{6.29}$$

Substitute $W = W_z$ and $Z = \Delta W$ into (6.29) and combine with (6.27). If $\Delta W \neq 0$, divide by $\|\Delta W\|_F$; if $\Delta W = 0$, the conclusion is immediate. This proves (6.24). □

Corollary 6.10 (Stability and strict active feasibility of the primal candidate)**.** *Let* $t_\star := R - \operatorname{tr} X_\star > 0$ *and* $\sigma_\star := \lambda_{\min}(\Sigma_\star) > 0$*. Define*

$$\begin{aligned}R_{\mathrm{aux}} &:= \min\left\{R_1, \frac{\sigma_\star}{2(1 + L_W)}, \frac{t_\star}{2\sqrt{s}(1 + L_W)}\right\},\\ X(y) &:= U(y)W(y)U(y)^\top, \qquad L_X := L_W + 2RL_U.\end{aligned} \tag{6.30}$$

For $y \in \overline{B}(y_\star, R_{\mathrm{aux}})$*,*

$$W(y) \succeq \frac{\sigma_\star}{2}I_s, \qquad \operatorname{tr} W(y) \le R - \frac{t_\star}{2}. \tag{6.31}$$

Moreover, $X(y)$ *is the unique solution of* $\min_{X \in \mathcal{D}_s(y)} \mathcal{J}_y(X)$*,* $\operatorname{rank} X(y) = s$*,* $X(y_\star) = X_\star$*, and*

$$\|X(y) - X(z)\|_F \le L_X\|y - z\| \qquad \big(y, z \in \overline{B}(y_\star, R_{\mathrm{aux}})\big). \tag{6.32}$$

Proof. Theorem 6.9 gives $\|W(y)-\Sigma_\star\|_F \le L_W R_{\mathrm{aux}}$. Weyl's inequality and $|\operatorname{tr}(W(y)-\Sigma_\star)| \le \sqrt{s}\|W(y)-\Sigma_\star\|_F$ therefore give (6.31). The map $W \mapsto U(y)WU(y)^\top$ is a bijection from Θ_s to $\mathcal{D}_s(y)$, with inverse $X \mapsto U(y)^\top XU(y)$, and preserves the objective in (6.21). This proves uniqueness of the auxiliary primal solution. It also preserves rank and gives $X(y_\star) = U_\star\Sigma_\star U_\star^\top = X_\star$. Finally, expand

$$\begin{aligned} X(y)-X(z) &= U(y)(W(y)-W(z))U(y)^\top \\ &\quad + (U(y)-U(z))W(z)U(y)^\top + U(z)W(z)(U(y)-U(z))^\top. \end{aligned}$$

Since $\|W(z)\|_F \le R$, the three terms have total norm at most $(L_W + 2RL_U)\|y-z\|$, proving (6.32). □

Define the auxiliary dual update

$$\Phi(y) := y + \rho\big(\mathcal{A}(X(y)) - b\big), \qquad L_\Phi := 1 + \rho\|\mathcal{A}\|L_X. \tag{6.33}$$

Then $\Phi(y_\star) = y_\star$ and $\|\Phi(y)-\Phi(z)\| \le L_\Phi\|y-z\|$ on $\overline{B}(y_\star, R_{\mathrm{aux}})$. This map is defined through the auxiliary problem before any comparison with the actual DD-QP.

Lemma 6.11 (Zero stationarity on the entire active block)**.** *For every $y \in \overline{B}(y_\star, R_{\mathrm{aux}})$,*

$$U(y)^\top S(\Phi(y))U(y) = 0, \qquad \langle S(\Phi(y)), X(y)\rangle = 0. \tag{6.34}$$

The compressed matrix is also zero in every other orthonormal frame of range $P_s(y)$*. Furthermore, for every $z \in \mathcal{Y}^* \cap \overline{B}(y_\star, R_{\mathrm{aux}})$,*

$$P_s(z) = P_\star, \qquad X(z) = X_\star, \qquad \Phi(z) = z. \tag{6.35}$$

Proof. By (6.31), $W(y)$ is positive definite with trace strictly less than R. Thus for every $Z \in \mathbb{S}^s$, both $W(y)+tZ$ and $W(y)-tZ$ belong to Θ_s for sufficiently small $t > 0$. The derivative in every direction must vanish. Substitution of (6.33) into (6.25) yields

$$0 = Dq_y(W(y))[Z] = -\langle U(y)^\top S(\Phi(y))U(y), Z\rangle \qquad (Z \in \mathbb{S}^s).$$

Taking $Z = U(y)^\top S(\Phi(y))U(y)$ proves the first identity in (6.34); taking its inner product with $W(y)$ proves the second. Any other orthonormal frame of the same space is $U(y)O$ for an orthogonal O, so its compression is $O^\top 0 O = 0$.

For a dual optimum z in the stated ball, Lemma 6.5 gives $\ker S(z) = \operatorname{range} X_\star$ and hence $P_s(z) = P_\star$. Therefore $\langle S(z), X\rangle = 0$ for every $X \in \mathcal{D}_s(z)$, so $\mathcal{J}_z(X) \ge 0$ on this set. The feasible matrix $X_\star$ attains zero, since $\mathcal{A}(X_\star) = b$. Corollary 6.10 gives $X(z) = X_\star$; the definition of Φ then gives $\Phi(z) = z$. □

6.3 Local exactness of the DD correction

We now compare the actual DD subproblem with the auxiliary problem on the current active space. The radii below control the current dual variable and the previous primal matrix jointly; no convergence of an algorithmic sequence is assumed.

Theorem 6.12 (Unique matrix exactness in a joint neighborhood)**.** *Suppose Assumption 2.1 holds, $X_\star$ is the unique primal optimum, and $1 \le s = \operatorname{rank} X_\star < n$. Fix $\rho > 0$ and*

$$s \le r \le n, \qquad 0 < \varepsilon \le \frac{\sigma_\star}{4\sqrt{r}}. \tag{6.36}$$

There exist $a_y \in (0, R_{\mathrm{aux}})$ and $a_X > 0$ such that, whenever

$$\|y - y_\star\| \le a_y, \qquad X^- \in \mathcal{D}, \quad \|X^- - X_\star\|_F \le a_X, \qquad V \in \operatorname{TopEig}_r(S(y)), \tag{6.37}$$

the reverse Cholesky factor $L = \mathrm{chol}(V^\top X^- V + \varepsilon I_r)$ *satisfies*

$$\operatorname*{argmin}_{X \in \mathcal{K}(V,L)} \mathcal{J}_y(X) = \{X(y)\} = \operatorname*{argmin}_{X \in \mathcal{D}_s(y)} \mathcal{J}_y(X). \tag{6.38}$$

In particular, the primal output has rank s *and the dual update is* $\Phi(y) = y + \rho(\mathcal{A}(X(y)) - b)$*. The statement is uniform over every admissible spectral basis* V*; uniqueness is asserted for the matrix output, not for its atomic coefficients.*

The proof uses two conditions: the auxiliary solution must remain feasible in the DD model, and atoms involving additional directions must have strictly negative support at $\Phi(y)$. We first express these conditions through continuous maps.

Definition 6.13 (Active coordinates and normalized atoms)**.** Fix $\varepsilon > 0$. For $X^- \in \mathcal{D}$ and $V \in \mathrm{St}(n,r)$, write

$$V = [V_A, V_I], \qquad L = \mathrm{chol}(V^\top X^- V + \varepsilon I_r) = \begin{pmatrix} L_A & L_{AI} \\ 0 & L_I \end{pmatrix}, \tag{6.39}$$

where V_A has s columns and L_A is $s \times s$. Define the continuous maps

$$\begin{aligned} T_A^\varepsilon : B(y_\star, R_{\mathrm{aux}}) \times \mathcal{D} \times \mathrm{St}(n,r) &\longrightarrow \mathbb{S}^s, \\ T_A^\varepsilon(y, X^-, V) &= L_A^{-1} V_A^\top X(y) V_A L_A^{-\top}, \end{aligned} \tag{6.40}$$

$$\begin{aligned} a_z^\varepsilon : \mathcal{D} \times \mathrm{St}(n,r) &\longrightarrow \mathbb{R}^n, \\ a_z^\varepsilon(X^-, V) &= \frac{VLz}{\|VLz\|}, \qquad z \in \mathcal{V}_r. \end{aligned} \tag{6.41}$$

Thus $A_z = R a_z^\varepsilon (a_z^\varepsilon)^\top$ is the trace-R matrix atom from Proposition 2.4. Partition the direction set as

$$\mathcal{V}_A = \{z \in \mathcal{V}_r : z_{s+1:r} = 0\}, \qquad \mathcal{V}_I = \mathcal{V}_r \setminus \mathcal{V}_A. \tag{6.42}$$

The second set is empty when $r = s$.

The stated continuity holds on the full displayed domains, without a spectral-basis restriction. Indeed, $X(y)$ is continuous, the matrix $V^\top X^- V + \varepsilon I_r$ is positive definite, and reverse Cholesky factorization is continuous. Its leading triangular block is invertible, so block extraction and inversion preserve continuity. Moreover,

$$\|VLz\| = \|Lz\| \geq \sqrt{\varepsilon}\|z\| > 0, \tag{6.43}$$

which establishes continuity of every normalized atom.

Lemma 6.14 (A sufficient certificate for the DD correction)**.** *Let* $(y, X^-, V) \in B(y_\star, R_{\mathrm{aux}}) \times \mathcal{D} \times \mathrm{St}(n,r)$ *and* $\varepsilon > 0$*. Suppose*

$$\begin{aligned} V \in \mathrm{TopEig}_r(S(y)), \qquad & T_A^\varepsilon(y, X^-, V) \in \mathsf{DD}_s, \\ (a_z^\varepsilon)^\top S(\Phi(y)) a_z^\varepsilon \leq 0 \quad & \textit{for every } z \in \mathcal{V}_I. \end{aligned} \tag{6.44}$$

Then $X(y) \in \operatorname{argmin}_{X \in \mathcal{K}(V,L)} \mathcal{J}_y(X)$*, where* $L = \mathrm{chol}(V^\top X^- V + \varepsilon I_r)$*. If the last inequality is strict for every* $z \in \mathcal{V}_I$*, then*

$$\operatorname*{argmin}_{X \in \mathcal{K}(V,L)} \mathcal{J}_y(X) = \{X(y)\}. \tag{6.45}$$

When $r = s$*, the strict-sign condition is empty and the minimizer is unique.*

Proof. Put $T = \mathrm{diag}(T_A^\varepsilon, 0) \in \mathsf{DD}_r$. The upper triangular structure in (6.39) gives

$$\begin{aligned} VLTL^\top V^\top &= V_A L_A T_A^\varepsilon L_A^\top V_A^\top \\ &= V_A V_A^\top X(y) V_A V_A^\top = X(y), \end{aligned}$$

because $V_A V_A^\top = P_s(y)$ and $X(y) \in \mathcal{D}_s(y)$. Since $\operatorname{tr} X(y) < R$ by Corollary 6.10, the candidate is feasible in $\mathcal{K}(V, L)$, even if $L_{AI} \neq 0$.

For any $X \in \mathcal{K}(V, L)$, expanding the squared residual in $\mathcal{J}_y$ gives the exact identity

$$\begin{aligned}\mathcal{J}_y(X) - \mathcal{J}_y(X(y)) &= \frac{\rho}{2}\|\mathcal{A}(X - X(y))\|^2 + \langle \rho\mathcal{A}^*(\mathcal{A}(X(y)) - b) - S(y), X - X(y)\rangle \\ &= \frac{\rho}{2}\|\mathcal{A}(X - X(y))\|^2 - \langle S(\Phi(y)), X - X(y)\rangle. \end{aligned} \tag{6.46}$$

By Proposition 2.4, write

$$X = \sum_{z \in \mathcal{V}_r} \alpha_z A_z, \qquad \alpha_z \geq 0, \qquad \sum_z \alpha_z \leq 1. \tag{6.47}$$

For $z \in \mathcal{V}_A$, triangularity gives $VLz = V_A L_A z_A \in \operatorname{range} P_s(y)$. Lemma 6.11 therefore yields

$$(a_z^\varepsilon)^\top S(\Phi(y)) a_z^\varepsilon = 0, \qquad \langle S(\Phi(y)), X(y)\rangle = 0. \tag{6.48}$$

The sign condition for the other atoms and (6.46) now imply $\mathcal{J}_y(X) \geq \mathcal{J}_y(X(y))$, proving optimality.

If all the inactive signs are strict, equality forces both $\mathcal{A}(X - X(y)) = 0$ and $\alpha_z = 0$ for every $z \in \mathcal{V}_I$. Consequently $X - X(y) = U(y) Z U(y)^\top$ for a symmetric Z. The uniform injectivity bound from Lemma 6.8 then gives

$$0 = \|\mathcal{A}(X - X(y))\| \geq \frac{\alpha_\star}{2}\|Z\|_F,$$

so $X = X(y)$. The same argument applies directly when $r = s$. □

Remark 6.15. The weak sign in (6.44) certifies optimality. The strict sign used below also excludes inactive atoms from any other optimal representation, and thereby certifies matrix uniqueness.

Lemma 6.16 (Uniform margins at the optimal parameters)**.** *For every $V \in \operatorname{TopEig}_r(S(y_\star))$ and every $0 < \varepsilon \leq \sigma_\star/(4\sqrt{r})$, set*

$$T_A^\star = T_A^\varepsilon(y_\star, X_\star, V), \qquad a_z^\star(\varepsilon) = a_z^\varepsilon(X_\star, V), \qquad \kappa_\varepsilon = \frac{\gamma_\star \varepsilon}{4(R + \varepsilon)} > 0.$$

Then

$$(T_A^\star)_{ii} - \sum_{j \neq i} |(T_A^\star)_{ij}| \geq \frac{3}{4}, \qquad 1 \leq i \leq s, \tag{6.49}$$

$$(a_z^\star(\varepsilon))^\top S(y_\star) a_z^\star(\varepsilon) \leq -2\kappa_\varepsilon, \qquad z \in \mathcal{V}_I. \tag{6.50}$$

The constants in these bounds do not depend on the choice of V.

Proof. The active gap implies that V_A spans range $X_\star$ and V_I lies in its orthogonal complement. Hence

$$V^\top X_\star V = \begin{pmatrix} W_A^\star & 0 \\ 0 & 0 \end{pmatrix}, \qquad W_A^\star = V_A^\top X_\star V_A \succeq \sigma_\star I_s.$$

Uniqueness of the reverse Cholesky factor gives

$$L = \begin{pmatrix} L_A & 0 \\ 0 & \sqrt{\varepsilon} I_{r-s} \end{pmatrix}, \qquad L_A L_A^\top = W_A^\star + \varepsilon I_s. \tag{6.51}$$

It follows that

$$T_A^\star = I_s - \varepsilon L_A^{-1} L_A^{-\top}, \qquad \|T_A^\star - I_s\|_{\mathrm{op}} \leq \frac{\varepsilon}{\sigma_\star + \varepsilon}. \tag{6.52}$$

Here the norm bound uses the smallest singular value of L_A; it does not interchange the factors in $L_A^{-1}L_A^{-\top}$. For any symmetric E, Cauchy–Schwarz on its ith row gives $\sum_j |E_{ij}| \le \sqrt{s}\|E\|_{\mathrm{op}}$. Applying this to $E = T_A^\star - I_s$ yields

$$(T_A^\star)_{ii} - \sum_{j\neq i} |(T_A^\star)_{ij}| \ge 1 - \frac{\sqrt{s}\,\varepsilon}{\sigma_\star + \varepsilon} > 1 - \frac{\sqrt{s}\,\varepsilon}{\sigma_\star} \ge \frac{3}{4}.$$

If $z \in \mathcal{V}_I$, the directions e_i and $(e_i \pm e_j)/\sqrt{2}$ satisfy $\|z_I\|^2 \ge \|z\|^2/2$. Equation (6.51) shows that the inactive component of VLz is $\sqrt{\varepsilon}V_I z_I$. Since $S(y_\star)$ vanishes on the active space and is bounded above by $-\gamma_\star I$ on its orthogonal complement, while $\|Lz\|^2 \le (\operatorname{tr} X_\star + \varepsilon)\|z\|^2 \le (R+\varepsilon)\|z\|^2$, we obtain

$$(a_z^\star(\varepsilon))^\top S(y_\star) a_z^\star(\varepsilon) \le -\frac{\gamma_\star \varepsilon \|z_I\|^2}{\|Lz\|^2} \le -\frac{\gamma_\star \varepsilon}{2(R+\varepsilon)} = -2\kappa_\varepsilon.$$

□

Lemma 6.17 (Uniform active DD margins and inactive atom negativity)**.** *Fix ε as in* (6.36) *and put $\kappa = \kappa_\varepsilon$. There exist $a_y \in (0, R_{\mathrm{aux}})$ and $a_X > 0$ such that every triple in* (6.37) *satisfies*

$$(T_A^\varepsilon)_{ii} - \sum_{j\neq i} |(T_A^\varepsilon)_{ij}| \ge \frac{1}{2}, \qquad 1 \le i \le s, \tag{6.53}$$

$$(a_z^\varepsilon)^\top S(\Phi(y)) a_z^\varepsilon \le -\kappa, \qquad z \in \mathcal{V}_I. \tag{6.54}$$

For every $z \in \mathcal{V}_A$, the support value is exactly zero.

Proof. The maps in Definition 6.13 are continuous, as are Φ and S. Thus each row margin in (6.53) and each support value in (6.54) is a continuous scalar function of (y, X^-, V). Moreover, $\Phi(y_\star) = y_\star$ because $X(y_\star) = X_\star$ and $\mathcal{A}(X_\star) = b$.

We verify that this continuity yields a neighborhood common to all admissible spectral bases. If no such neighborhood existed, we could choose triples (y_j, X_j^-, V_j) with

$$y_j \longrightarrow y_\star, \qquad X_j^- \longrightarrow X_\star, \qquad V_j \in \mathrm{TopEig}_r(S(y_j)),$$

for which at least one required inequality fails. The Stiefel manifold is compact, so a subsequence satisfies $V_j \to V_\infty \in \mathrm{St}(n,r)$. The ordered spectral-basis map has a closed graph: passing to the limit in $S(y_j)v_{j,i} = \lambda_i(S(y_j))v_{j,i}$, using continuity of eigenvalues, gives $V_\infty \in \mathrm{TopEig}_r(S(y_\star))$. Since there are finitely many rows and directions, a further subsequence violates the same row bound or the same atom bound. In the first case, continuity gives a limiting margin at most $1/2$, contradicting (6.49). In the second case it gives a limiting support value at least $-\kappa$, contradicting (6.50). This proves the existence of positive joint radii. Shrinking them if necessary places their closed balls inside the domain of the auxiliary solution.

Finally, for an active direction z, triangularity implies $VLz \in \operatorname{range} P_s(y)$. Its zero support value follows exactly from Lemma 6.11, rather than from continuity. □

Proof of Theorem 6.12. Choose the radii supplied by Lemma 6.17. Equation (6.53) ensures $T_A^\varepsilon \in \mathsf{DD}_s$, and (6.54) supplies the strict signs required in Lemma 6.14. Consequently the actual DD problem has the unique solution $X(y)$. Corollary 6.10 identifies the same matrix as the unique auxiliary minimizer and gives $\operatorname{rank} X(y) = s$. Substitution into the dual update yields $\Phi(y)$. All bounds hold uniformly over the admissible spectral bases, which proves the theorem. □

6.4 Second-order model accuracy and linear convergence

Consider a triple (y, X^-, V) in the joint neighborhood of Theorem 6.12, and set $L = \mathrm{chol}(V^\top X^- V + \varepsilon I_r)$. The actual DD model is

$$m_{\mathcal{K}(V,L)}(z) := b^\top z + \max_{X \in \mathcal{K}(V,L)} \langle S(z), X\rangle. \tag{6.55}$$

For V_s consisting of the first s columns of V, the auxiliary active-space model is

$$m_{V_s}(z) := b^\top z + \max_{X \in \mathcal{D}_s(y)} \langle S(z), X \rangle = b^\top z + R[\lambda_1(V_s^\top S(z) V_s)]_+. \tag{6.56}$$

Both feasible sets are compact convex subsets of $\mathcal{D}$. Theorem 6.12 shows that their correction problems share the unique primal minimizer $X(y)$. Applying Lemma 4.1 separately to $\mathcal{K}(V, L)$ and $\mathcal{D}_s(y)$ gives

$$\begin{aligned} \Phi(y) &= \operatorname*{argmin}_{z \in \mathbb{R}^m} \left\{ m_{\mathcal{K}(V,L)}(z) + \frac{\|z - y\|^2}{2\rho} \right\} \\ &= \operatorname*{argmin}_{z \in \mathbb{R}^m} \left\{ m_{V_s}(z) + \frac{\|z - y\|^2}{2\rho} \right\} = y + \rho\big(\mathcal{A}(X(y)) - b\big). \end{aligned} \tag{6.57}$$

The support identity in that lemma, applied separately to each set, shows that $X(y)$ attains both support maxima at $\Phi(y)$. Therefore

$$m_{\mathcal{K}(V,L)}(\Phi(y)) = m_{V_s}(\Phi(y)) = b^\top \Phi(y) + \langle S(\Phi(y)), X(y) \rangle. \tag{6.58}$$

Thus a model-error estimate for the auxiliary spectral model at $\Phi(y)$ also holds for the actual DD model at its computed update.

Lemma 6.18 (Hidden Ritz gap controls the spectral-model error)**.** *Let R_Y satisfy (6.6). For every $y \in \overline{B}(y_\star, R_Y)$, every $z \in \mathbb{R}^m$, and $V_s \in \mathrm{TopEig}_s(S(y))$,*

$$0 \le F(z) - m_{V_s}(z) \le L_{\mathrm{sp}} \|z - y\|^2, \qquad L_{\mathrm{sp}} = \frac{2R\|\mathcal{A}\|^2}{\gamma_\star}. \tag{6.59}$$

In particular, this is the constant defined in (2.47), independent of the chosen ordered frame, ρ, and ε.

Proof. First suppose also $z \in \overline{B}(y_\star, R_Y)$. Let $\lambda_1 \ge \cdots \ge \lambda_n$ be the eigenvalues of $S(z)$, and let $\theta_1 \ge \cdots \ge \theta_s$ be those of $V_s^\top S(z) V_s$. Rayleigh–Ritz gives $\theta_i \le \lambda_i$ for $1 \le i \le s$. Since the positive-part map is nondecreasing and 1-Lipschitz,

$$\begin{aligned} 0 \le F(z) - m_{V_s}(z) &= R\big([\lambda_1]_+ - [\theta_1]_+\big) \le R(\lambda_1 - \theta_1) \\ &\le R \sum_{i=1}^{s} (\lambda_i - \theta_i) = R\langle S(z), P_s(z) - P_s(y) \rangle. \end{aligned}$$

The last equality uses $V_s V_s^\top = P_s(y)$ and the Ky Fan characterization of $P_s(z)$. The same characterization at y implies $\langle S(y), P_s(z) - P_s(y) \rangle \le 0$. Hence

$$\begin{aligned} R\langle S(z), P_s(z) - P_s(y) \rangle &\le R\langle S(z) - S(y), P_s(z) - P_s(y) \rangle \\ &\le R\|S(z) - S(y)\|_F \|P_s(z) - P_s(y)\|_F \\ &\le R\|\mathcal{A}\| L_P \|z - y\|^2 \le L_{\mathrm{sp}} \|z - y\|^2. \end{aligned}$$

The last inequality follows from $L_P = \sqrt{2}\|\mathcal{A}\|/(7\gamma_\star/8) \le 2\|\mathcal{A}\|/\gamma_\star$. This is the hidden Ritz-gap argument specialized to an exact leading s-dimensional space: the center Ritz gap is zero, leaving only the quadratic projector-motion term.

For completeness, a direct Rayleigh–Ritz comparison extends the same bound to arbitrary evaluation points $z \in \mathbb{R}^m$. Write $G = S(y)$, $E = S(z) - S(y)$, $P = P_s(y)$, and $g = \lambda_s(G) - \lambda_{s+1}(G) \ge 7\gamma_\star/8$. Let u be a unit leading eigenvector of $G + E$ and set $t = \|(I - P)u\|^2$. If $Pu \ne 0$, let $w = Pu/\|Pu\|$; otherwise choose any unit vector $w \in \mathrm{range}\, P$. In either case,

$$(u^\top w)^2 = 1 - t, \qquad \|uu^\top - ww^\top\|_F = \sqrt{2t}.$$

Since P is a spectral projector of G, its cross blocks vanish, and

$$\begin{aligned} u^\top Gu - w^\top Gw &= ((I-P)u)^\top G((I-P)u) - t\, w^\top Gw \\ &\le t\lambda_{s+1}(G) - t\lambda_s(G) = -gt. \end{aligned}$$

Also $\lambda_1(V_s^\top S(z)V_s) \ge w^\top S(z)w$. Consequently,

$$\begin{aligned} 0 &\le \lambda_1(S(z)) - \lambda_1(V_s^\top S(z)V_s) \\ &\le u^\top Gu - w^\top Gw + \langle E, uu^\top - ww^\top\rangle \\ &\le -gt + \sqrt{2t}\|E\|_F \le \frac{\|E\|_F^2}{2g}. \end{aligned}$$

The last step completes the square in $\sqrt{t}$. Applying the positive-part map and using $\|E\|_F \le \|\mathcal{A}\|\|z-y\|$ gives

$$0 \le F(z) - m_{V_s}(z) \le \frac{R\|\mathcal{A}\|^2}{2g}\|z-y\|^2 \le L_{\mathrm{sp}}\|z-y\|^2,$$

which proves the asserted extension. □

Combining (6.58) with Lemma 6.18 gives

$$0 \le F(\Phi(y)) - m_{\mathcal{K}(V,L)}(\Phi(y)) \le L_{\mathrm{sp}}\|\Phi(y)-y\|^2 \tag{6.60}$$

whenever the model center y lies in the spectral neighborhood and local exactness holds. This is precisely the quadratic model-error condition in Lemma 4.6. In the proof below we also control the trial point directly, before using the distance decrease.

Proof of Theorem 2.5. All constants below depend only on the fixed problem data, the selected optimal pair, and the fixed parameters r, ρ, ε. Let $a_y, a_X > 0$ be the joint-neighborhood radii from Theorem 6.12, and let $R_{\mathrm{aux}}, L_X, L_\Phi$ be the auxiliary-solution radius and Lipschitz constants from Corollary 6.10 and (6.33). Enlarge L_X, L_Φ to be at least 1, and choose

$$\delta_X := \frac{a_X}{2}, \qquad \delta_y := \min\left\{\frac{a_y}{2}, \frac{R_{\mathrm{aux}}}{4}, \frac{R_Y}{2L_\Phi}, \frac{r_{\mathrm{QG}}}{2}, \frac{\delta_X}{L_X}\right\} > 0. \tag{6.61}$$

These radii are fixed before the algorithm starts.

First we establish the invariant joint neighborhood. Suppose the kth state satisfies $\|y_k - y_\star\| \le \delta_y$ and $X_k \in \mathcal{D}$ with $\|X_k - X_\star\|_F \le \delta_X$. Theorem 6.12 gives the unique QP matrix $X_{k+1} = X(y_k)$ and the update $y_{k+1} = \Phi(y_k)$. Before using any model-error or invariance conclusion, the independent Lipschitz estimate for Φ and $\Phi(y_\star) = y_\star$ give

$$\|\Phi(y_k) - y_\star\| \le L_\Phi\|y_k - y_\star\| \le R_Y/2.$$

Both y_k and its computed update therefore lie in the spectral neighborhood. Since $2\rho L_{\mathrm{sp}} < 1$, (6.57), (6.60), and the distance conclusion of Lemma 4.6 give

$$\|y_{k+1} - y^o\| \le \|y_k - y^o\| \qquad \text{for every } y^o \in \mathcal{Y}^*. \tag{6.62}$$

Taking $y^o = y_\star$ yields $\|y_{k+1} - y_\star\| \le \delta_y$. The auxiliary primal map similarly satisfies

$$\|X_{k+1} - X_\star\|_F = \|X(y_k) - X(y_\star)\|_F \le L_X\|y_k - y_\star\| \le \delta_X, \qquad X_{k+1} \in \mathcal{D}.$$

The initialization starts this induction. It proves (i), uniqueness of every QP matrix output, and well-definedness of all iterates.

The invariant neighborhood lies within the quadratic-growth ball, so we may now use the contraction and objective conclusions of Theorem 4.7 at every iteration:

$$d_{k+1}^2 \le \frac{1}{1+\rho\mu}d_k^2, \qquad F(y_{k+1}) - f^* \le \frac{d_k^2}{2\rho}.$$

With $q = (1 + \rho\mu)^{-1/2} < 1$, induction gives

$$d_{k+1} \le q d_k, \qquad d_k \le d_0 q^k, \qquad F(y_{k+1}) - f^* \le \frac{d_0^2}{2\rho} q^{2k}.$$

This proves (ii) and the claimed dual objective estimate.

Let $z_k = \mathrm{proj}_{\mathcal{Y}^*}(y_k)$. Equation (6.62) gives $\|y_{k+1} - z_k\| \le d_k$, and hence

$$\|y_{k+1} - y_k\| \le \|y_{k+1} - z_k\| + \|y_k - z_k\| \le 2d_k \le 2d_0 q^k. \tag{6.63}$$

The right side of (6.63) is summable, so (y_k) is Cauchy and has a limit y_∞. Since $d_k \to 0$ and $\mathcal{Y}^*$ is closed, $y_\infty \in \mathcal{Y}^*$. Summing the step bounds from k onwards gives

$$\|y_k - y_\infty\| \le \sum_{j=k}^{\infty} \|y_{j+1} - y_j\| \le \frac{2d_0}{1-q} q^k,$$

which proves (iii).

It remains to transfer these estimates to the primal outputs. Since

$$\|z_k - y_\star\| \le d_k + \|y_k - y_\star\| \le 2\delta_y \le R_{\mathrm{aux}}/2,$$

the point z_k lies in the auxiliary-solution neighborhood, and (6.35) gives $X(z_k) = X_\star$. Lipschitz continuity of the auxiliary map now yields

$$\|X_{k+1} - X_\star\|_F = \|X(y_k) - X(z_k)\|_F \le L_X \|y_k - z_k\| = L_X d_k \le L_X d_0 q^k.$$

Finally, the dual update and (6.63) imply

$$\|\mathcal{A}(X_{k+1}) - b\| = \frac{\|y_{k+1} - y_k\|}{\rho} \le \frac{2d_0}{\rho} q^k.$$

This proves the remaining primal and residual estimates. □

6.5 Extension to joint multi-cone corrections

The product extension follows the same sequence of tasks, but one point requires care. Exact-block variables need not be unique, so the single-block argument cannot simply be applied to a unique complete primal matrix. We instead compare the *sets* of joint minimizers. The assumption that all primal optima share the same large blocks provides the stability needed for their DD representations, while uniqueness of the image under $\mathcal{A}$ provides a well-defined dual update.

Proof of Theorem 3.1(ii). *Stability of the joint auxiliary correction.* Use the standing strict-radius assumptions and Assumption 3.1. Let $P_j(y)$ be the leading rank-s_j projector of $S_j(y)$ and define

$$\mathcal{D}_{\mathrm{act}}(y) := \prod_{i \in \mathcal{E}} \mathcal{D}_i \times \prod_{j \in \mathcal{L}} \{X_j \in \mathcal{D}_j : X_j = P_j(y) X_j P_j(y)\}. \tag{6.64}$$

The solution set $\mathcal{X}_{\mathrm{act}}(y) := \mathrm{argmin}_{X \in \mathcal{D}_{\mathrm{act}}(y)} \mathcal{J}_y(X)$ is nonempty by compactness. By Lemma 4.1, every minimizer $\bar{X}$ induces the proximal minimizer of the same active model. That proximal minimizer is unique, because its objective is strongly convex. Consequently, all auxiliary minimizers have the same image $\mathcal{A}\bar{X}$, and $\Phi(y) := y + \rho(\mathcal{A}\bar{X} - b)$ is well defined for any $\bar{X} \in \mathcal{X}_{\mathrm{act}}(y)$.

Retain $x_{i,\star}$ and replace every large optimal block by $P_j(y) X_{j,\star} P_j(y)$. This comparison point is auxiliary-feasible, since projection preserves positive semidefiniteness and does not increase the trace. As

$y \to y_\star$, it tends to $X_\star$ and its correction objective tends to zero. Any subsequential limit $\bar{X}$ of auxiliary minimizers has its large blocks supported in $\ker S_j(y_\star)$ and satisfies

$$\mathcal{J}_{y_\star}(\bar{X}) = \frac{\rho}{2}\|\mathcal{A}\bar{X} - b\|^2 - \sum_{i\in\mathcal{E}}\langle S_i(y_\star), \bar{x}_i\rangle \geq 0.$$

Every term is nonnegative by dual feasibility. The feasible comparison forces the sum to vanish, so the limit is feasible and complementary, and hence primal optimal. The common-optimal-large-block assumption then identifies its large blocks as $X_{j,\star}$. A compactness and subsequence argument makes this conclusion uniform over all auxiliary minimizers:

$$\sup_{\bar{X}\in\mathcal{X}_{\mathrm{act}}(y)} \|\bar{X}_j - X_{j,\star}\|_F \longrightarrow 0, \qquad \Phi(y) \longrightarrow y_\star \qquad (y \to y_\star). \tag{6.65}$$

Thus every auxiliary minimizer has positive definite large-block active compressions and strict trace bounds near $y_\star$. These are the interior properties required to apply the blockwise exactness certificate.

Verification of the joint exactness certificate. At every such auxiliary minimizer, first-order optimality gives

$$\begin{aligned} P_j(y)S_j(\Phi(y))P_j(y) = 0 \qquad & (j \in \mathcal{L}), \\ \bar{x}_i \in \operatorname*{argmax}_{x\in\mathcal{D}_i}\langle S_i(\Phi(y)), x\rangle \qquad & (i \in \mathcal{E}). \end{aligned} \tag{6.66}$$

The first identity uses both signs of small symmetric active-block perturbations. The optimal-state calculations (6.52)–(6.50) apply to each large block. For fixed ε_j, they provide strict DD representability and strictly negative support of every additional atom. Uniform auxiliary stability, continuity of reverse Cholesky, and the frame compactness argument in Lemma 6.17 extend these margins to one joint neighborhood, uniformly over all auxiliary minimizers. Hence each auxiliary solution is DD-feasible.

For a DD-feasible X and an auxiliary minimizer $\bar{X}$, expand

$$\mathcal{J}_y(X) - \mathcal{J}_y(\bar{X}) = \frac{\rho}{2}\|\mathcal{A}(X - \bar{X})\|^2 - \sum_{\ell}\langle S_\ell(\Phi(y)), X_\ell - \bar{X}_\ell\rangle. \tag{6.67}$$

For an exact block, support optimality makes its contribution nonnegative. For a large block, the active contribution is zero, and a positive coefficient on any additional atom makes the contribution strictly positive. Hence every auxiliary minimizer is DD-optimal. Conversely, equality at a DD minimizer forces all additional atoms to vanish. That minimizer therefore lies in the full active product domain and attains its optimal value. This proves equality of the joint minimizer sets, without requiring injectivity of the complete active product map or uniqueness of the exact-block variables.

Model accuracy at the common update. Let the columns of $V_{j,A}$ form an orthonormal basis of range $P_j(y)$ and define

$$m_y^{\mathrm{act}}(v) := b^\top v + \sum_{i\in\mathcal{E}}\sigma_{\mathcal{D}_i}(S_i(v)) + \sum_{j\in\mathcal{L}} R_j[\lambda_{\max}(V_{j,A}^\top S_j(v)V_{j,A})]_+. \tag{6.68}$$

Choose a common radius $\bar{R}_Y > 0$ such that $\|\mathcal{A}_j\|\bar{R}_Y \leq \gamma_j/16$ for every $j \in \mathcal{L}$. Applying the Rayleigh–Ritz estimate in the proof of Lemma 6.18 to each large block, and noting that the exact-block support terms cancel, gives

$$0 \leq F(v) - m_y^{\mathrm{act}}(v) \leq 2\sum_{j\in\mathcal{L}}\frac{R_j\|\mathcal{A}_j\|^2}{\gamma_j}\|v - y\|^2 = L_{\mathrm{mix}}\|v - y\|^2$$

for all $y \in \overline{B}(y_\star, \bar{R}_Y)$ and $v \in \mathbb{R}^m$. Take any common minimizer $\bar{X}$ of the joint DD and active corrections. Applying the support identity in Lemma 4.1 to each model gives

$$m_y^{\mathrm{DD}}(\Phi(y)) = b^\top\Phi(y) + \langle S(\Phi(y)), \bar{X}\rangle = m_y^{\mathrm{act}}(\Phi(y)),$$

where the inner product denotes the sum over all blocks. Thus the actual joint DD update satisfies

$$0 \le F(\Phi(y)) - m_y^{\mathrm{DD}}(\Phi(y)) \le L_{\mathrm{mix}}\|\Phi(y) - y\|^2. \tag{6.69}$$

As in the single-block proof, this estimate applies to the actual trial without a separate preliminary restriction on $\Phi(y)$. It can therefore be used to establish invariance next.

Joint neighborhood invariance. Let joint exactness hold for previous large-block errors at most $a_X > 0$. By (6.65), choose a sufficiently small dual ball on which every auxiliary output has large-block errors at most $a_X/2$. Place the ball inside the center exactness, spectral-gap, and quadratic-growth regions, and initialize the previous large blocks within $a_X/2$. The actual correction is auxiliary-optimal, so (6.69) applies. Lemma 4.6 and $2\rho L_{\mathrm{mix}} < 1$ keep the new center in the same ball; auxiliary stability keeps its large blocks within $a_X/2$. These two conclusions close the induction and prove joint invariance. No neighborhood restriction on the exact-block variables is needed beyond membership in their compact sets.

Convergence and identified ranks. From the switching index k_{sw} onward, apply Theorem 4.7 with $L = L_{\mathrm{mix}}$ and the growth modulus μ of the complete penalty. Write $d_{\mathrm{sw}} := \mathrm{dist}(y_{k_{\mathrm{sw}}}, \mathcal{Y}^*)$ and $q := (1+\rho\mu)^{-1/2}$. For every $t \ge 0$,

$$\begin{aligned} \mathrm{dist}(y_{k_{\mathrm{sw}}+t}, \mathcal{Y}^*) &\le d_{\mathrm{sw}} q^t, \\ F(y_{k_{\mathrm{sw}}+t+1}) - f^* &\le \frac{d_{\mathrm{sw}}^2}{2\rho} q^{2t}, \\ \|\mathcal{A}X_{k_{\mathrm{sw}}+t+1} - b\| &\le \frac{2d_{\mathrm{sw}}}{\rho} q^t. \end{aligned} \tag{6.70}$$

The dual sequence converges R-linearly to an optimum. For the primal outputs, the exact support identity gives

$$\langle C, X_{k+1}\rangle - f^* = h_{k+1} - e_k + \langle y_{k+1}, \mathcal{A}X_{k+1} - b\rangle.$$

The dual, residual, and quadratic model-error estimates make the right-hand side tend to zero. Thus every primal cluster point is feasible and optimal. Compactness and the common-optimal-large-block assumption imply $X_{j,k} \to X_{j,\star}$. Moreover, joint exactness and positive definite active compressions give rank s_j for every new large-block output in the neighborhood. This proves convergence of those blocks, but not a matrix-distance rate: such a rate would require additional solution-map stability.

Total model work. Each DD iteration uses one joint correction. For $0 < \epsilon < d_{\mathrm{sw}}$,

$$N_{\mathrm{total}}(\epsilon) \le N_{\mathrm{pre}} + \left\lceil \frac{\log(d_{\mathrm{sw}}/\epsilon)}{\log(1/q)} \right\rceil = N_{\mathrm{pre}} + O(\log(1/\epsilon)). \tag{6.71}$$

The prefix counts every diagonal correction, including rejected trials. No DD correction is needed for this dual-distance target when $d_{\mathrm{sw}} \le \epsilon$. Each work unit is an exact joint convex quadratic conic correction; numerical inner iterations and spectral computations are excluded. □

This completes the connection between the product construction and the common convergence theorem. In Algorithm 3.1, the switching state (X_k, y_k) uses the same indexing as the single-block algorithm. The guarantee applies when this state enters the joint neighborhood; finiteness of the warm start alone does not establish that fact. Exact blocks introduce no model approximation error, but their effect on the complete penalty and the joint solve remains part of the analysis.

7 Numerical experiments

We conduct three numerical studies. The first compares the cost of solving DD atom-QPs and full semidefinite correction subproblems, examining the computational benefit of replacing PSD constraints by polyhedral coefficient constraints. The second compares five primal correction mechanisms on a random MaxCut SDP, with particular attention to the tradeoff between diagonal, DD, and full reduced semidefinite models. The third reports a separate comparison between a PDHG implementation and MOSEK on six application-derived SDPs; it does not evaluate the diagonal–DD hybrid.

7.1 Computational cost of DD-QP and SDP corrections

A principal motivation for the DD model is to reduce the cost of the coupled correction subproblem. We first examine this benefit on six synthetic instances with block order $N \in \{10, 50\}$ and number of blocks $K \in \{1, 10, 100\}$. Each instance has $m = 200$ random linear measurements coupling all K blocks. The trace bound on each block and the quadratic-penalty parameter are both set to one. For each pair (N, K), we generate one instance using a fixed random seed and compare the full PSD-constrained subproblem with its DD atomic restriction.

The DD atom-QP is solved by a safeguarded Barzilai–Borwein (BB) projected-gradient method, implemented on both CPU and GPU. For the full SDP, we use a GPU implementation of the BB method, with NVIDIA cuSOLVER batched eigendecompositions for the PSD projections, and COPT on CPU. All BB runs start from zero, use double precision, and terminate based on independently evaluated relative primal–dual gap and feasibility checks with tolerance 10^{-9}. COPT's internal tolerances are also set to 10^{-9}. GPU computations use an NVIDIA RTX 4090, while CPU computations use a single thread on an Intel Core Ultra 9 185H.

Table 2 reports the median solve time over three repetitions on each fixed instance. The timings include stopping-certificate evaluations but exclude setup, data transfer, and warm-up. Thus, they measure the cost of solving an already constructed correction subproblem rather than the complete cost of an outer iteration.

Within the GPU BB implementations, the DD-QP has a lower solve time than the full SDP on all six instances. The difference is particularly pronounced for $N = 50$, where the speedup ranges from approximately 7.2 to 13.7. For $(N, K) = (50, 100)$, the DD-QP takes 0.00615 seconds, compared with 0.04403 seconds for the SDP. This observation is consistent with the simpler projections and optimality checks of the atomic QP, which avoid the eigendecompositions required by the SDP implementation.

The CPU comparison also shows substantial savings on the largest instance: BB solves the DD-QP in 0.32909 seconds, whereas COPT solves the SDP in 2.95216 seconds, a time ratio of approximately 9. Comparing GPU BB for the DD-QP with CPU COPT for the SDP gives a ratio of approximately 480 on this instance. The latter comparison combines differences in formulation, numerical method, and hardware, rather than measuring the effect of the DD formulation alone. CPU methods remain competitive on smaller instances: CPU COPT is fastest for $(N, K) = (10, 1)$, and CPU BB is fastest for $(N, K) = (10, 10)$.

These results support the DD atomic formulation as an inexpensive correction model for the tested family, particularly for the larger block order. The comparison concerns subproblem solution cost, however, rather than equality of the resulting corrections. The DD-QP uses an inner approximation of the PSD feasible set and need not attain the same optimal objective value as the full SDP. Accordingly, the numerical tolerances assess accuracy for each respective subproblem; this experiment does not test the local exactness property established in the preceding analysis.

7.2 Comparison of primal update models

We first consider a weighted MaxCut problem with $n = 20$ vertices. Each undirected edge is sampled with probability 0.3, and a cycle is added to ensure connectivity. The edge weights are uniform on $[0.5, 1.5)$. The seed 20260929 produces a graph with 72 edges. Let L be its weighted Laplacian. With the maximization convention of (2.1), the semidefinite relaxation is

$$\max_{X \succeq 0} \langle C, X \rangle \quad \text{subject to} \quad \mathcal{A}X = b, \qquad C = \frac{1}{4}L, \quad \mathcal{A}X = \operatorname{diag}(X), \quad b = \mathbf{1}. \tag{7.1}$$

The formulation contains one PSD block of order 20, 210 independent scalar matrix entries, and 20 equality constraints.

The five mechanisms are the single-atom, diagonal-QP, triangular DD-QP, reduced-SDP, and truncated-PDHG updates. To compare the first four within a common iteration, we initialize them at $X_0 = \mathbf{1}\mathbf{1}^\top$ and

Table 2: Median solve times in seconds over three repetitions. Timings include stopping-certificate evaluations and exclude setup, data transfer, and warm-up. GPU methods use an NVIDIA RTX 4090; CPU methods use one thread on an Intel Core Ultra 9 185H. DD-QP uses the DD atomic restriction, whereas SDP retains the full PSD feasible set. The fastest method for each instance is indicated in bold.

		$N = 10$			$N = 50$		
Model	Method	$K = 1$	$K = 10$	$K = 100$	$K = 1$	$K = 10$	$K = 100$
DD-QP	GPU BB	0.050	0.044	**0.022**	**0.006**	**0.006**	**0.006**
	CPU BB	0.028	**0.032**	0.048	0.008	0.029	0.329
SDP	GPU BB (batched)	0.059	0.063	0.045	0.085	0.045	0.044
	CPU COPT	**0.018**	0.045	0.332	0.035	0.256	2.952

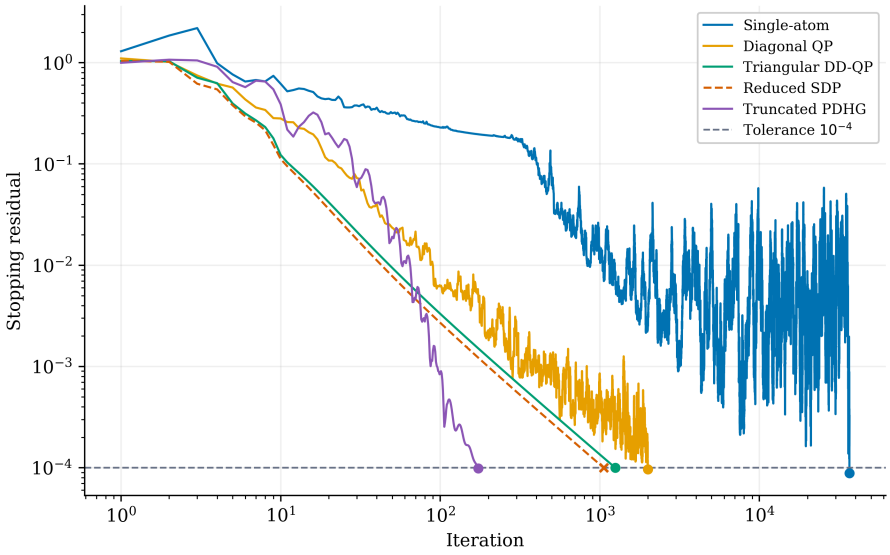


Figure 1: Residual histories for five primal update mechanisms on a random MaxCut SDP with 20 vertices and 72 edges. Markers indicate the first iteration satisfying the corresponding stopping tolerance of 10^{-4}. All residuals are recorded at every iteration without smoothing or historical-minimum filtering. Truncated PDHG uses a different stopping measure from the other four methods.

$y_0 = 0$ and use

$$\begin{aligned} X_{k+1} &\in \operatorname*{argmin}_{X \in \mathcal{K}_k} \left\{ -\langle S(y_k), X \rangle + \frac{\rho_k}{2} \|\mathcal{A}X - b\|^2 \right\}, \\ y_{k+1} &= y_k + \rho_k(\mathcal{A}X_{k+1} - b), \qquad \rho_k = \frac{1}{k+1}, \quad k \geq 0. \end{aligned} \tag{7.2}$$

For these four methods, the correction models impose $\operatorname{tr}(X) = 20$, as required by feasibility in (7.1). The parameter sequence is fixed in advance, and both the objective and multiplier update follow the maximization convention of Section 2.1.

The single-atom model is the segment from X_k to $20 v_k v_k^\top$, where v_k is a leading eigenvector of $S(y_k) = C - \mathcal{A}^* y_k$. The diagonal model instead jointly optimizes the nonnegative weights of X_k and three trace-20 atoms from the leading three eigenvectors, with the weights summing to one. The triangular DD model has working dimension $r = 3$ and uses the nine atoms obtained from reverse Cholesky factorization. Its regularization on the normalized primal matrix is $\varepsilon = 10^{-3}$, and its atomic weights are nonnegative with sum one. Finally, the reduced SDP uses

$$\mathcal{K}_k = \left\{ \beta X_k + 20 V_k W V_k^\top : W \succeq 0,\ \beta \geq 0,\ \beta + \operatorname{tr}(W) = 1 \right\},$$

where V_k contains the leading three eigenvectors of $S(y_k) = C - \mathcal{A}^* y_k$.

Truncated PDHG uses $X_0 = 0$, $y_0 = 0$, fixed step sizes $\tau = \sigma = 0.9$, and extrapolation parameter $\theta = 1$. Its projection retains at most three positive eigenvalues, giving $\operatorname{rank}(X_k) \leq 3$, and it has no explicit trace constraint. All five mechanisms use fixed working dimensions, without restarts or acceptance tests.

These choices delimit the purpose of the experiment. The varying penalty, trace-equality normalization, omitted acceptance tests, and uncertified DD initialization differ from the algorithms and hypotheses analyzed in Sections 5 and 6. We therefore use this study to compare correction mechanisms, not to verify the complexity theorems for Algorithms 2.1–3.1. The reported iterations are experimental outer updates rather than accepted steps or total model solves in the theoretical analysis.

Clarabel solves the QP and reduced-SDP subproblems with requested tolerance 10^{-11} and at most 200 inner iterations. A separate model-gap check uses tolerance 10^{-7}. The QP objectives are scaled equivalently and have constant terms removed; the reduced SDP retains its original objective representation. Runs use single-threaded BLAS, and timings are taken after warm-up.

Each method stops at the first iteration at which its stopping residual is at most 10^{-4}, subject to limits of 300000 iterations and 180 seconds. For the first four methods, the dual gap is evaluated after recovering the trace multiplier. Truncated PDHG instead uses the original SDP's dual residual and gap. Thus its stopping measure is not identical to that of the other four methods.

The residual histories are associated with Figure 1. Among the mechanisms sharing (7.2), the reported iteration counts are 36605 for the single-atom update, 1999 for diagonal QP, 1248 for triangular DD-QP, and 1056 for the reduced SDP. The reported DD trajectory closely follows the reduced-SDP trajectory, whereas the single-atom and diagonal trajectories have more pronounced oscillations near the stopping threshold. On this instance, the DD model's additional within-subspace freedom reduces the iteration count relative to the diagonal model while retaining a QP correction.

Truncated PDHG reaches its own stopping threshold in 173 iterations. This number should not be interpreted as an isolated effect of the primal model: the initialization, outer update, and stopping measure are also different. Moreover, iteration counts do not measure the different costs of the corrections. The experiment therefore illustrates the mechanisms on one instance, rather than establishing a general ranking by runtime or validating the hybrid's convergence guarantees.

7.3 Experiments on application-derived SDP instances

We next report a separate solver comparison on six SDP instances from the ComputationalRobotics `pendulum` repository.[1] The instances form three groups, VL, FR, and CR, each with $N \in \{5, 10\}$. Presolve is applied before numerical solution. This study concerns the reported PDHG implementation and MOSEK; it does not compare the diagonal and DD correction algorithms.

Table 3: Problem dimensions after presolve and runtimes on application-derived SDP instances. Here, q_{PSD} is the number of PSD blocks, $n_{\max}$ is the largest block order, n_{var} is the number of independent scalar decision variables, and m is the number of equality constraints. Runtimes are in seconds. TL indicates that the termination criterion was not met within the 1200-second time limit. The lower completed runtime in each row is bold.

	Problem dimensions				Runtime (s)	
Problem	q_{PSD}	$n_{\max}$	n_{var}	m	PDHG	MOSEK
VL $N = 5$	54	141	49022	39530	**24.44**	430.63
VL $N = 10$	109	143	107733	86253	**32.99**	808.11
FR $N = 5$	59	177	71956	55731	**48.48**	1166.51
FR $N = 10$	119	176	160997	123684	**76.18**	TL
CR $N = 5$	114	185	95782	71272	**97.90**	348.70
CR $N = 10$	229	185	201837	149447	**210.40**	698.32

[1] https://github.com/ComputationalRobotics/pendulum

Table 3 gives the dimensions after presolve and the measured runtimes. Here q_{PSD} is the number of PSD blocks, $n_{\max}$ is their largest order, and the number of independent scalar matrix variables is

$$n_{\mathrm{var}} = \sum_{j=1}^{q_{\mathrm{PSD}}} \frac{n_j(n_j+1)}{2},$$

where n_j is the order of the jth PSD block; m denotes the number of equality constraints. The resulting problems contain between 54 and 229 PSD blocks, with largest block orders between 141 and 185. Their scalar variable counts range from 49022 to 201837, and their equality constraint counts range from 39530 to 149447.

The reported PDHG runtimes are lower on all six instances. For the five completed MOSEK runs, the ratio of MOSEK runtime to PDHG runtime is approximately 3.3–24.5. On FR with $N = 10$, PDHG completes in 76.18 seconds, while MOSEK does not meet the termination criterion within 1200 seconds. All six PDHG runs complete in less than 211 seconds.

Within each family, increasing N primarily enlarges the number of blocks and the coupled equality system, rather than the largest PSD block. The measurements therefore concern problems with many coupled PSD blocks. They show lower observed runtimes for the reported PDHG implementation, but provide no direct timing evidence for the diagonal-to-DD hybrid. The experimental record also lacks complete hardware, software-version, and common accuracy settings. These details are needed before the table can support a reproducible solver-performance comparison.

8 Concluding remarks

In this paper, we developed a convex framework that separates spectral subspace selection from the representation used for a low-rank SDP correction. This perspective connects several existing correction mechanisms and leads to two realizations whose large-block coefficient constraints are polyhedral. The diagonal method controls model error by refinement and enjoys global finite-accuracy guarantees. The primal-adapted DD method uses a richer coefficient model and, under the stated local assumptions, reproduces the full active-space semidefinite correction. This exactness is the key step that permits local linear convergence without solving that auxiliary SDP.

The common support-model analysis separates three issues that must be distinguished in assessing these algorithms: progress per accepted center update, the model refinements needed to obtain that update, and the numerical work required to solve each correction. The multi-cone extension preserves this distinction. Exact blocks add no model approximation error, but remain part of the coupled optimization problem and of the geometry of its complete penalty.

A A sufficient condition for bounded relevant sublevels

For completeness, we verify the bounded-multiplier condition used for the uniform raw objective estimate in Theorem 4.5. The result below applies to the single-block trace-bounded penalty. This additional condition is not needed for the averaged guarantees, the dual and residual bounds once growth is available, or the finite-target argument in Section 5.2.

Lemma A.1 (Primal Slater feasibility and bounded relevant sublevels)**.** *Under the standing assumptions for (2.1), suppose there exists $X^{\mathrm{sf}} \succ 0$ with $\mathcal{A}X^{\mathrm{sf}} = b$. Then $\mathcal{E}_0 \cap \mathcal{L}_H$ is bounded for every finite $H \geq 0$, where $\mathcal{E}_0 = y_0 + \operatorname{range}\mathcal{A}$ and $\mathcal{L}_H = \{y : F(y) - f^* \leq H\}$.*

Proof. We first construct a strictly feasible point inside the trace cap. Let X^* be the optimum satisfying $\operatorname{tr} X^* < R$. For sufficiently small $t > 0$, the convex combination $X^{\mathrm{s}} := (1-t)X^* + tX^{\mathrm{sf}}$ satisfies $X^{\mathrm{s}} \succ 0$, $\mathcal{A}X^{\mathrm{s}} = b$, and $\operatorname{tr} X^{\mathrm{s}} < R$. Set $\vartheta := \frac{1}{2}\min\{\lambda_{\min}(X^{\mathrm{s}}), (R - \operatorname{tr} X^{\mathrm{s}})/\sqrt{n}\} > 0$. If $\|Z\|_F \leq \vartheta$, the inequalities

$\|Z\|_{\mathrm{op}} \le \|Z\|_F$ and $|\operatorname{tr} Z| \le \sqrt{n}\|Z\|_F$ imply $X^{\mathrm{s}} + Z \in \mathcal{D}$. Maximizing the Lagrangian over this Frobenius ball gives

$$F(y) \ge b^\top y + \langle C - \mathcal{A}^* y, X^{\mathrm{s}} \rangle + \vartheta \|C - \mathcal{A}^* y\|_F = \langle C, X^{\mathrm{s}} \rangle + \vartheta \|C - \mathcal{A}^* y\|_F. \tag{A.1}$$

Therefore, for $y \in \mathcal{L}_H$,

$$\|\mathcal{A}^* y\|_F \le \|C\|_F + \frac{f^* + H - \langle C, X^{\mathrm{s}} \rangle}{\vartheta}.$$

This bounds $\mathcal{A}^* y$, but redundant multiplier directions still need to be removed. Let $\mathcal{S} := \operatorname{range} \mathcal{A}$ and $y_\perp := \Pi_{\ker \mathcal{A}^*} y_0$. Every $y \in \mathcal{E}_0$ has the unique representation $y = y_\perp + v$ with $v \in \mathcal{S}$, and $\mathcal{A}^* y = \mathcal{A}^* v$. If $\mathcal{S} = \{0\}$, then $\mathcal{E}_0 = \{y_0\}$ is already bounded. Otherwise, $\ker \mathcal{A}^* = \mathcal{S}^\perp$ makes $\mathcal{A}^*$ injective on $\mathcal{S}$. Compactness of its unit sphere gives a positive minimum norm $\sigma_{\mathcal{A}}$. It follows that

$$\|y\| \le \|y_\perp\| + \|v\| \le \|y_\perp\| + \frac{1}{\sigma_{\mathcal{A}}} \left(\|C\|_F + \frac{f^* + H - \langle C, X^{\mathrm{s}} \rangle}{\vartheta} \right) \qquad (y \in \mathcal{E}_0 \cap \mathcal{L}_H).$$

This bound is uniform on the relevant sublevel and proves the claim. □

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